Intelligent Air Quality Forecasting and Risk Alert System

PROJECT REPORT

Submitted to

BHARATHIDASAN UNIVERSITY, TIRUCHIRAPPALLI-24

in partial fulfillment of the requirements for the award of

BACHELOR OF SCIENCE IN DATA SCIENCE

Submitted by

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SHRIMATI INDIRA GANDHI COLLEGE

AFFILIATED TO BHARATHIDASAN UNIVERSITY
(NATIONALLY ACCREDITED AT “A” GRADE (4TH CYCLE BY NAAC)
AN ISO 9001-2015 CERTIFIED INSTITUTION
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APRIL – 2026

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This is to certify that this project entitled "**INTELLIGENT AIR QUALITY FORECASTING AND RISK ALERT SYSTEM**" is a bonafide record of the work done and

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in partial fulfillment of the requirements for the award of the degree **BACHELOR OF SCIENCE IN DATA SCIENCE** of **Bharathidasan University, Tiruchirappalli** during the sixth semester of the academic year April-2026. This is the original work of the candidates and was done under my guidance and supervision.

Signature of the Head of the Department

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Signature of the Guide

VIVA-VOCE

VIVA-VOCE EXAMINATION

The project VIVA-VOCE Examination is held

on 13-03-2026

13/3/26 13/3/2026

Signature of the Internal Examiner

Signature of the External Examiner

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ACKNOWLEDGEMENT

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ABSTRACT

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Air pollution has become a major environmental and public health concern, requiring accurate prediction and timely alerts to minimize its impact on urban populations. This project presents an intelligent city-wise air quality prediction and short-term forecasting system using machine learning techniques. The proposed system predicts the current Air Quality Index (AQI) and forecasts the air quality level for the next 24 hours based on multi-pollutant indicators, including carbon monoxide (CO), ozone (O_3), nitrogen dioxide (NO_2), and particulate matter (PM2.5). The model learns temporal pollution patterns from historical data and performs 24-hour ahead AQI category prediction for different cities. In addition to forecasting, the system incorporates a risk-based alert mechanism that automatically generates warnings when the predicted air quality reaches severe or hazardous levels. Visual analytics in the form of graphs are provided to help users understand pollution trends and city-level air quality variations. By combining real-time assessment, short-term forecasting, alert generation, and city-wise analysis, the proposed system supports proactive air quality monitoring and decision-making. This approach can assist individuals, health agencies, and urban planners in taking preventive measures against air pollution risks.

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INTRODUCTION

1.Introduction

Air pollution has become one of the most critical environmental challenges worldwide, directly affecting human health, climate conditions, ecosystems, and overall quality of life. Rapid industrialization, urbanization, increased vehicular emissions, and energy consumption have significantly contributed to the deterioration of air quality in many regions. Poor air quality is associated with severe health issues such as respiratory diseases, cardiovascular disorders, asthma, lung cancer, and premature mortality. Therefore, monitoring and predicting air quality has become essential for sustainable urban planning and public health protection. The Air Quality Prediction System is designed to forecast the Air Quality Index (AQI) for the next 24 hours using historical environmental data and machine learning techniques. The Air Quality Index is a standardized indicator used to communicate how polluted the air currently is or how polluted it is forecasted to become. It typically considers major pollutants such as PM2.5 (Particulate Matter $\leq 2.5\mu m$), PM10, CO (Carbon Monoxide), NO_2 (Nitrogen Dioxide), SO_2 (Sulphur Dioxide), and O_3 (Ozone). These pollutants are measured through environmental monitoring stations and are influenced by various meteorological factors including temperature, humidity, wind speed, and atmospheric pressure. This project leverages data-driven approaches to analyse patterns in historical air pollution levels and meteorological conditions. By applying time-series analysis and predictive modeling techniques such as regression models, decision trees, random forests, or deep learning algorithms (e.g., LSTM networks), the system identifies trends, seasonal patterns, and correlations between pollutants and weather variables. The trained model then generates predictions for the next 24-hour AQI levels. The primary objectives of this project are:

- To analyse historical air quality data and identify pollutant trends.
- To develop a predictive model capable of forecasting air quality for the next 24 hours.

- To evaluate model performance using appropriate metrics such as RMSE, MAE, or R^2.
- To provide actionable insights that can help individuals, policymakers, and environmental agencies take preventive measures.

The system architecture typically involves data collection, preprocessing (handling missing values, normalization, feature engineering), model training, validation, and deployment. Proper preprocessing ensures data consistency and improves model accuracy. Feature engineering may include lag features, rolling averages, and time-based features (hour, day, season), which are particularly important in time-series forecasting. By forecasting air quality in advance, this system can help

- Issue early health advisories to sensitive populations.
- Support government authorities in implementing traffic control or emission reduction strategies.
- Assist industries in optimizing operations to reduce pollutant release.
- Raise public awareness about environmental conditions.

Need, Scope, and Motivation for Air Quality Prediction System

Air pollution levels fluctuate continuously due to various factors such as traffic density, industrial activities, weather conditions, and seasonal changes. Sudden increases in pollution levels often occur without sufficient warning, leaving individuals and authorities unprepared. Vulnerable populations, including children, the elderly, and people with respiratory conditions, are particularly affected by poor air quality. Short-term air quality prediction plays a crucial role in issuing early health advisories, planning outdoor activities and transportation, supporting hospital preparedness, enabling timely pollution control measures, and improving public awareness and safety. A 24-hour prediction window is especially important because it provides sufficient time for authorities and citizens to take preventive actions while still maintaining reasonable prediction accuracy.

The scope of this project is focused on short-term air quality prediction using historical data. It considers a specific geographic region, selected air pollutants and meteorological parameters, and involves offline model training and prediction. The project does not include real-time deployment or long-term climate modeling however, the system is designed to be extensible and can be enhanced in the future. The motivation behind this project arises from the increasing frequency of air pollution episodes and their direct impact on daily life. Many urban areas experience severe air quality degradation, particularly during certain seasons. Despite the availability of monitoring systems, the lack of reliable short-term prediction limits effective response. This project aims to bridge the gap between data availability and actionable insights by transforming historical air quality data into meaningful future predictions. It demonstrates how data science can be applied to address critical environmental and public health challenges

SYSTEM ANALYSIS

2.SYSTEM ANALYSIS

2.1 Existing Model

Overview of the Existing System and Traditional Models

The existing air quality monitoring systems are primarily designed to measure and report the current concentration of pollutants in the atmosphere. These systems rely on physical monitoring stations installed in various geographical locations that continuously collect pollutant data such as PM2.5, PM10, CO, NO_2, SO_2, and O_3. The collected data is processed to calculate the Air Quality Index (AQI), which indicates whether the air quality is categorized as Good, Moderate, Unhealthy, or Hazardous. While these systems are effective in providing real-time information, they are generally limited in their ability to predict future air quality conditions, especially short-term forecasts such as the next 24 hours. Most traditional systems focus more on monitoring rather than forecasting. In earlier approaches, air quality prediction was performed using classical statistical methods such as Linear Regression, ARIMA (AutoRegressive Integrated Moving Average), and Moving Average models. Linear regression attempts to establish a linear relationship between pollutant concentration and influencing factors like temperature, humidity, and wind speed; however, since air pollution is affected by highly complex and nonlinear relationships, it often fails to capture sudden spikes or seasonal variations. ARIMA, a widely used time-series forecasting model, predicts future values based on past observations and performs reasonably well for stable datasets. Nevertheless, it assumes linear relationships, requires stationary data, struggles with sudden environmental changes, and does not effectively handle multiple influencing variables. Moving Average models smooth short-term fluctuations and highlight trends, but they are not suitable for accurate short-term air quality prediction because they primarily rely on historical averages rather than dynamic environmental interactions.

Limitations of the Existing Model and Need for Improvement

The existing air quality monitoring systems and traditional prediction models suffer from several key limitations. Most government and environmental systems primarily focus on real-time monitoring and provide only the current AQI values rather than predictive insights. Traditional statistical models often assume linearity and stationarity, which do not accurately represent real-world atmospheric conditions, leading to limited forecasting accuracy. Air quality depends on complex interactions between traffic density, industrial emissions, weather conditions, seasonal patterns, and geographical factors; however, many existing models fail to capture these nonlinear relationships effectively. Additionally, short-term air quality prediction requires understanding time-series dependencies such as hourly trends, daily cycles, and seasonal effects, but basic models struggle with capturing long-range temporal patterns. Environmental datasets frequently contain missing values and noisy or irregular readings, and many traditional approaches do not handle such inconsistencies efficiently. Furthermore, some systems require manual intervention and lack automation and scalability, making them unsuitable for large-scale, real-time forecasting. Due to these limitations, there is a strong need for an improved predictive system that can accurately forecast air quality for the next 24 hours, effectively handle nonlinear relationships, incorporate meteorological parameters, learn temporal patterns from historical data, and provide early warnings for public safety. Advanced machine learning or deep learning-based models can address these challenges by identifying complex patterns in large datasets and generating more reliable and accurate forecasts.

2.3 Proposed System

Overview of the Proposed System

The proposed Air Quality Prediction System is a data-driven, machine learning-based framework developed to estimate both the current Air Quality Index (AQI) and the next 24-hour AQI using environmental and pollutant parameters. The system is designed to transform raw air pollution data into meaningful predictions and health advisories, enabling proactive environmental monitoring and decision-making. The system operates by collecting pollutant concentration levels such as PM2.5, PM10,

NO_2, SO_2, CO, and O_3, along with meteorological variables including temperature, humidity, and wind speed. These parameters serve as input features to the trained machine learning model. Before model training, the dataset undergoes preprocessing steps such as data cleaning, handling missing values, and feature preparation to ensure consistency and reliability. A Random Forest Regression model is employed to predict the continuous AQI value. This algorithm is selected due to its ability to model nonlinear relationships and handle complex environmental interactions effectively. The predicted AQI value is then categorized into standard pollution levels such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe based on established AQI breakpoint ranges. This dual-output approach allows the system to function both as a regression model (predicting AQI value) and as a classification system (predicting AQI category). In addition to current AQI prediction, the system includes a short-term forecasting component that estimates air quality conditions for the next 24 hours. This forecasting capability enhances the system's practical usefulness by providing early warnings about potential pollution increases. The model's performance is evaluated using regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score, along with classification metrics including precision, recall, F1-score, and accuracy. The results indicate exceptionally high predictive performance, demonstrating the system's robustness and reliability. An interactive user interface developed using Gradio allows users to input environmental parameters and instantly obtain AQI predictions along with health advisory messages. This ensures that complex machine learning outputs are presented in a clear, accessible, and user-friendly format. Overall, the proposed system integrates data preprocessing, machine learning modeling, performance evaluation, and interactive deployment into a unified end-to-end air quality prediction framework.

Objectives of the Proposed System

The main objectives of the proposed system are:

- To develop a predictive model capable of forecasting air quality for the next 24 hours.
- To improve prediction accuracy compared to traditional statistical models.

- To analyse complex nonlinear relationships between pollutants and environmental factors.
- To provide timely alerts and insights for public health safety.
- To build a scalable and automated prediction framework.

Architecture of the Proposed Air Quality Prediction System

The architecture of the proposed Air Quality Prediction System is designed as a complete end-to-end data science pipeline that integrates data collection, preprocessing, model development, prediction, evaluation, forecasting, and user interface deployment into a structured framework. The system follows a modular architecture to ensure scalability, maintainability, and flexibility, allowing each component to function independently while contributing to the overall workflow. The Data Input Layer forms the foundation of the system and consists of historical air quality datasets containing pollutant concentrations such as PM2.5, PM10, NO_2, SO_2, CO, and O_3, along with meteorological variables including temperature, humidity, and wind speed. These variables act as independent features that influence the Air Quality Index (AQI). The collected dataset is used for both model training and evaluation. The Data Preprocessing and Feature Engineering Layer prepares the raw data for machine learning. In this stage, missing values and inconsistencies are handled, outliers are treated, and the data is normalized where necessary. Additional derived features may also be created to improve model performance. After preprocessing, the dataset is divided into training and testing sets to evaluate the model's ability to generalize to unseen data. The Model Training Layer uses the Random Forest Regression algorithm to capture complex relationships between pollutants, meteorological variables, and AQI values. Random Forest builds multiple decision trees using different subsets of data and combines their outputs to produce more accurate and stable predictions while reducing overfitting.

The Model Training Layer uses the Random Forest Regression algorithm to capture complex relationships between pollutants, meteorological variables, and AQI values. Random Forest builds multiple decision trees using different subsets of data and combines their outputs to produce more accurate and stable predictions while reducing overfitting. The Prediction and Evaluation Layer validates the model's performance using unseen test data. Regression metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score are used to measure prediction accuracy. The predicted AQI values are also mapped into standard AQI categories such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe, and evaluated using classification metrics including precision, recall, F1-score, and confusion matrix. The Forecasting Component enhances the system by predicting AQI levels for the next 24 hours, enabling short-term forecasting based on historical patterns. This supports proactive environmental monitoring and early warning for public health. Finally, the User Interface and Deployment Layer integrates the trained model into an interactive application developed using Gradio. Users can input pollutant and weather parameters, and the system displays the predicted AQI value, its category, and a corresponding health advisory. This interface makes the system accessible and easy to use for non-technical users. Overall, the modular architecture ensures accuracy, scalability, and practical deployment, making the system suitable for future enhancements such as real-time data integration and large-scale implementation.

PROBLEM STATEMENT

3.Problem Statement

Air pollution has become a critical environmental and public health concern in many urban and industrial regions. Rapid industrialization, increasing vehicular emissions, population growth, and environmental degradation have significantly contributed to rising pollution levels. Poor air quality leads to severe health problems such as respiratory diseases, cardiovascular issues, and reduced life expectancy. Therefore, accurate monitoring and prediction of Air Quality Index (AQI) are essential for early warning and preventive action. Traditional air quality monitoring systems primarily rely on manual reporting or static data dashboards that display historical pollutant concentrations. While these systems provide current air quality status, they often lack predictive capabilities and advanced data-driven analysis. Moreover, conventional methods may not effectively capture complex nonlinear relationships between multiple pollutants and meteorological factors that influence AQI levels. Although AQI is calculated using standard formulas based on pollutant concentrations, predicting future AQI trends requires intelligent modeling approaches capable of learning patterns from historical data. The absence of real-time predictive systems limits proactive decision-making by individuals, health authorities, and environmental agencies. Therefore, there is a need to develop a machine learning-based air quality prediction system that can accurately estimate current AQI values and forecast short-term future AQI levels using pollutant and environmental parameters. Such a system should not only provide numerical predictions but also categorize air quality levels and generate health advisory alerts to improve public awareness and preventive response. The objective of this project is to design and implement a data-driven Air Quality Prediction System that utilizes supervised machine learning techniques to model the relationship between pollutant concentrations and AQI levels. The system aims to provide accurate, reliable, and user-friendly air quality predictions to support environmental monitoring and public health management.

Problem Definition

The problem addressed in this project is the accurate prediction of Air Quality Index (AQI) values using environmental and pollutant concentration data. Air quality is influenced by multiple interacting factors such as PM2.5, PM10, NO_2, SO_2, CO, O_3, temperature, humidity, and wind speed, which exhibit complex and nonlinear relationships. Due to these dynamic interactions, traditional statistical methods are often insufficient for precise AQI forecasting. The objective of this project is to develop a supervised machine learning model capable of learning the relationship between pollutant concentrations and corresponding AQI values from historical data. The system is designed to perform two primary tasks: first, to predict the continuous AQI value as a regression problem; and second, to classify the predicted AQI into predefined pollution categories such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe as a classification problem. Given a set of pollutant and meteorological input features, the model should generate an accurate numerical AQI prediction and map it to the appropriate category using standard AQI breakpoint ranges. Additionally, the system aims to forecast short-term AQI values for the next 24 hours to provide early warnings and support proactive environmental management.

Objectives of the Project

The primary objective of this project is to develop a machine learning-based Air Quality Prediction System capable of accurately estimating current and short-term future AQI levels using pollutant and meteorological data. The project aims to collect and preprocess historical air quality and meteorological data, including PM2.5, PM10, NO_2, SO_2, CO, O_3, temperature, humidity, and wind speed, and perform data cleaning, missing value handling, normalization, and feature preparation to ensure quality and consistency. It also focuses on feature engineering by analysing relationships between environmental variables and creating meaningful features, including interaction terms and relevant transformations, to improve model performance. Another key objective is to develop a supervised regression model that can accurately predict continuous AQI values based on pollutant concentrations and meteorological parameters. Following regression, the predicted AQI values are categorized into standard pollution levels such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe using predefined AQI breakpoint ranges. The system further extends to short-term AQI forecasting for the next 24 hours by learning temporal patterns from historical data. Model performance is evaluated using regression metrics such as MAE, RMSE, and R^2 Score, and classification metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis to ensure reliability and robustness. Additionally, the project includes the development of an interactive user interface that allows users to input environmental parameters and receive AQI predictions along with health advisory messages. Finally, the system is designed to be scalable and user-friendly, supporting environmental monitoring, public awareness, and data-driven decision-making for air quality management.

Scope of the Project

The scope of this project focuses on analysing historical air quality and meteorological data to understand patterns, trends, and relationships affecting AQI levels. It involves the development of a machine learning-based system to predict continuous AQI values using supervised learning techniques based on pollutant concentrations and environmental factors. The project also includes the classification of predicted AQI values into standard pollution categories such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe according to predefined AQI ranges.

DATA COLLECTION

4.Data Collection

4.1 Data Description

Dataset Source

The dataset used in this project is obtained from the Kaggle repository titled Time Series Air Quality Data of India (2010–2023*)*. From the available monitoring station files, only AP001.csv was selected for analysis and model development.

Dataset Type

The dataset is a time-series air quality dataset containing historical records of pollutant concentrations and meteorological parameters collected from a specific monitoring station. The data is recorded at regular time intervals and includes both environmental and atmospheric measurements.

Attributes (Features)

The dataset consists of multiple input features representing pollutant concentrations and weather conditions. The major attributes include:

- PM2.5 (Particulate Matter ≤ 2.5 µm)
- PM10 (Particulate Matter ≤ 10 µm)
- NO_2 (Nitrogen Dioxide)
- SO_2 (Sulfur Dioxide)
- CO (Carbon Monoxide)
- O_3 (Ozone)
- Temperature
- Humidity
- Wind Speed
- Date and Time (Timestamp)

These features act as independent variables used to predict the Air Quality Index (AQI).

Target Variable

The target variable in this project is the Air Quality Index (AQI). The system performs:

- Regression: Predicting continuous AQI values
- Classification: Categorizing AQI into Good, Satisfactory, Moderate, Poor, Very Poor, and Severe

Data Characteristics

- The dataset is structured in CSV format.
- It contains time-stamped environmental observations.
- The data may contain missing values due to sensor downtime or measurement errors.
- The dataset includes both numerical and temporal features.

Data Preprocessing Requirements

Before model training, the dataset required:

- Handling missing values
- Removing irrelevant columns
- Converting date-time values into proper format
- Feature selection and preparation

Air Pollutant Attributes

Attribute	Description
PM2.5	Fine particulate matter with diameter ≤ 2.5 micrometres
PM10	Coarse particulate matter with diameter ≤ 10 micrometres
NO_2	Nitrogen dioxide concentration
SO_2	Sulphur dioxide concentration
CO	Carbon monoxide concentration
O_3	Ground-level ozone concentration

These pollutants are primary contributors to the Air Quality Index (AQI) and have significant health impacts.

4.2 Data Sources (Kaggle)

Introduction

Data sources play a vital role in the development of any machine learning–based prediction system. The reliability, diversity, and completeness of the dataset directly influence the accuracy and robustness of the model. In the context of air quality prediction, accessing large-scale, historical, and well-structured datasets is essential for capturing pollution trends, seasonal patterns, and the influence of meteorological factors. For this project, datasets were collected from Kaggle, a widely recognized online platform that hosts high-quality datasets shared by governments, research institutions, and data science communities. Kaggle provides open-access datasets that are commonly used for academic research, experimentation, and real-world machine learning applications. In modern data-driven projects, obtaining reliable and comprehensive datasets is often one of the biggest challenges. Kaggle addresses this challenge by offering structured and well-documented datasets that significantly reduce the time and effort required for data acquisition.

Overview of Kaggle as a Data Source

Kaggle is an online data science platform that offers:

- Publicly available datasets
- Verified and community-reviewed data
- Structured and well-documented files
- Support for real-world and academic projects

Kaggle datasets are particularly suitable for air quality prediction projects due to:

- Availability of long-term historical data
- Inclusion of multiple air pollutants and AQI values
- Integration of meteorological parameters

- Consistent data formats such as CSV

The datasets available on Kaggle are often compiled from official government agencies such as environmental monitoring boards, weather departments, and international environmental organizations.

Description of the Kaggle Air Quality Datasets

The air quality datasets used in this project consist of historical pollution measurements and environmental variables recorded over extended periods. These datasets are designed to support analysis, visualization, and predictive modeling of air pollution trends.

Nature of the Data

- Type: Time-series data
- Format: CSV (Comma-Separated Values)
- Granularity: Hourly or daily observations
- Coverage: Urban and metropolitan regions
- Domain: Environmental monitoring and public health

DATA CLEANING

5.Data Cleaning

Data cleaning is a crucial step in the data preprocessing phase, as raw environmental datasets often contain inconsistencies, missing values, duplicate records, and format variations. Proper cleaning ensures that the dataset is accurate, consistent, and suitable for reliable predictive modelling . The following steps were performed to enhance data quality before model development.

5.1 Removal of Duplicate Records

The dataset was examined for duplicate entries to prevent redundancy and biased learning during model training. Duplicate records can occur due to repeated data collection, system errors, or data merging processes. Such records may distort statistical calculations and negatively affect model performance. Therefore, duplicate rows were identified and removed to maintain data uniqueness and integrity. This step ensured that each observation represented a distinct environmental measurement.

5.2 Missing Value Analysis and Treatment

Environmental datasets, particularly those related to air quality monitoring, often contain missing values due to sensor failures, transmission errors, or incomplete data recording. A systematic approach was implemented to analyse and handle missing values effectively.

First, the percentage of missing values was calculated for each column (excluding date-related attributes). Based on this analysis:

- Columns with more than 50% missing values were removed from the dataset to avoid unreliable predictions and excessive imputation.
- For columns with acceptable levels of missing data (≤ 50%), a rule-based imputation strategy was applied
    - The 'Wind Direction (WD deg)' column was filled using the mode, as wind direction represents directional data where the most frequently occurring value preserves dominant wind patterns.

- All other numerical attributes were imputed using the median, which is less sensitive to extreme values and outliers commonly found in pollution measurements.

Date-related columns such as ‘From Date’ and ‘To Date’ were excluded from automatic imputation to preserve temporal consistency and were handled separately. This structured and automated missing value treatment improved data completeness while maintaining statistical reliability.

5.3 Correction of Inconsistent Data Formats

Raw datasets often contain inconsistencies in data types and formats, which can lead to errors during analysis and modeling. Therefore, format standardization was performed to ensure uniformity.

- **Date and Time Standardization**

  Date columns such as ‘From Date’ and ‘To Date’ were converted from string format to proper DateTime data types. This enabled chronological sorting, time-based filtering, and feature extraction such as year, month, day, and hour.

- **Categorical Data Standardization**

  Text-based categorical fields were examined for inconsistencies such as variations in capitalization, spelling differences, or extra spaces. These inconsistencies were corrected to maintain uniform representation across categories.

- **Unit Consistency**

  Environmental parameters were checked to ensure consistency in measurement units. Any discrepancies in units or representation were standardized to maintain analytical accuracy.

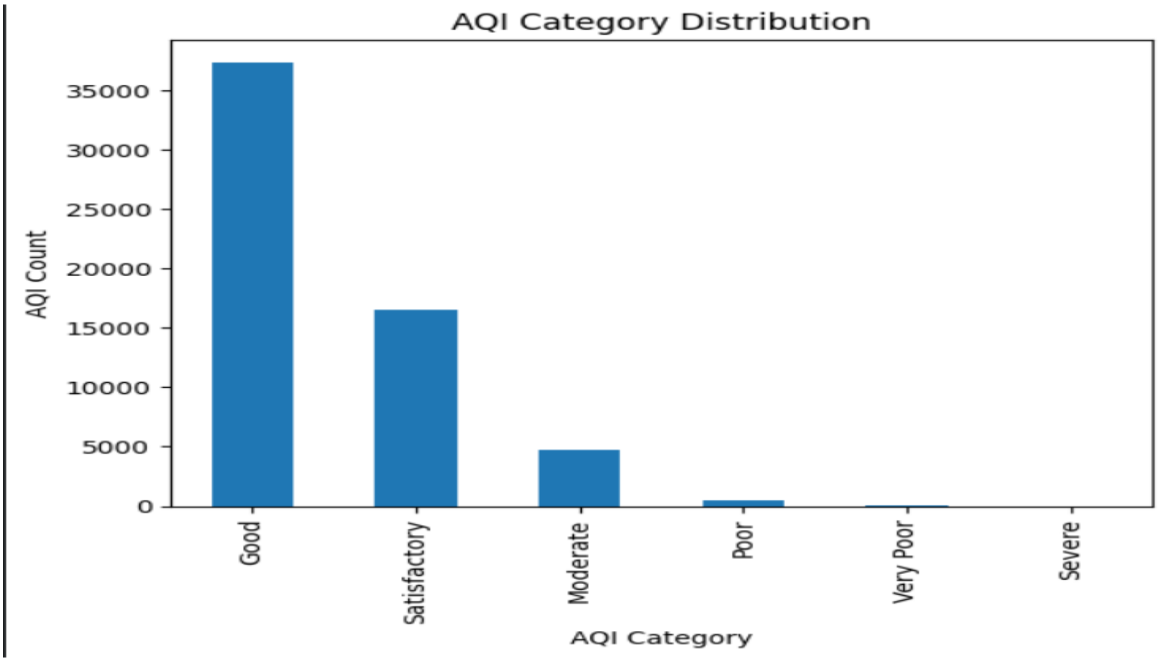
Handling Class Imbalance Using SMOTE

In classification problems, datasets often suffer from class imbalance, where one class (majority class) has significantly more samples than the other class (minority class). This imbalance can cause the machine learning model to become biased toward the majority class, leading to poor prediction performance for the minority class. To address this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied. SMOTE is an advanced data balancing technique that generates synthetic samples for the minority class rather than simply duplicating existing records. Unlike traditional oversampling methods that replicate minority samples, SMOTE creates new synthetic data points by interpolating between existing minority class samples. It selects a minority instance and identifies its nearest neighbours within the same class. Then, new synthetic samples are generated along the line segments connecting these neighbouring points. This approach improves the representation of the minority class in the training dataset, making the data distribution more balanced. As a result:

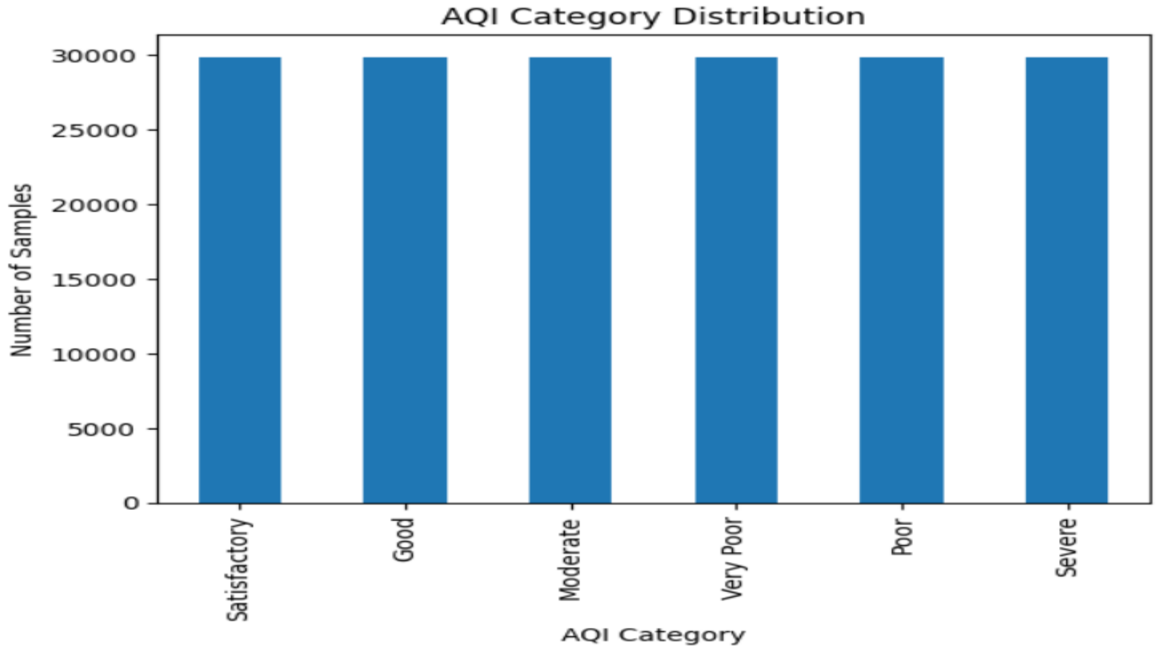
- The model learns patterns from both classes effectively.
- Bias toward the majority class is reduced.
- Recall and F1-score for the minority class improve.
- Overall classification performance becomes more reliable.

SMOTE was applied only to the training dataset after performing the train-test split to avoid data leakage. By balancing the class distribution, the training data quality was enhanced, leading to better generalization of the predictive model.

Before SMOTE



After SMOTE



AQI Category Distribution
30000
25000
20000
15000
10000
5000
0
Number of Samples
Satisfactory
Good
Moderate
Very Poor
Poor
Severe
AQI Category

SOFTWARE DESCRIPTION

6.Software Description

Python Programming Language

The proposed Air Quality Prediction System was developed using Python, a high-level, interpreted programming language widely adopted in scientific computing, artificial intelligence, and machine learning domains. Python was selected due to its simplicity, readability, modular structure, and extensive ecosystem of libraries that support data analysis, visualization, and predictive modeling. Python follows an object-oriented and procedural programming paradigm, enabling structured and efficient implementation of complex data processing tasks. In this project, Python was used for:

- Data acquisition and loading
- Data preprocessing and transformation
- Statistical and descriptive analysis
- Handling missing values and inconsistent formats
- Feature extraction and engineering
- Implementation of machine learning algorithms
- Model evaluation and validation
- Deployment through an interactive user interface

The flexibility of Python allowed seamless integration between preprocessing modules, modeling algorithms, and the user interface layer. Additionally, Python's large developer community ensures continuous updates and support for advanced data science frameworks.

Google Colaboratory (Cloud-Based Development Environment)

The entire project was implemented using Google Colaboratory (Google Colab), a cloud-hosted Jupiter notebook environment that allows execution of Python programs directly in a web browser. Google Colab was chosen for the following reasons:

- It eliminates the need for local installation and system configuration.
- Provides free access to computational resources.

- Supports real-time execution and visualization.
- Allows easy integration with Google Drive for dataset storage.
- Facilitates experimental model development in an interactive notebook format.

The notebook-based architecture of Google Colab enabled systematic documentation of each step in the machine learning workflow, including preprocessing, visualization, model training, and performance evaluation. This structured environment enhanced reproducibility and experimentation efficiency. Moreover, the cloud-based nature of Google Colab ensured that hardware limitations did not restrict computational tasks, making it an efficient platform for handling environmental datasets.

Gradio (Interactive User Interface Framework)

To convert the predictive model into a functional and user-accessible application, Gradio was integrated into the system. Gradio is an open-source Python framework that enables rapid development of web-based interfaces for machine learning models. Through Gradio, an interactive front-end was developed where users can input environmental parameters such as temperature, pollutant concentration, and atmospheric conditions. The system processes these inputs and provides real-time AQI predictions as output. The integration of Gradio provided several advantages:

- Simplified user interaction without requiring coding knowledge.
- Real-time model inference.
- Easy deployment capability.
- Web-based accessibility.

This transformation from a backend predictive model to an interactive application demonstrates the practical applicability of the proposed system in real-world environmental monitoring scenarios.

Overall Software Environment Justification

The combination of Python, Google Colab, specialized machine learning libraries, and Gradio created a comprehensive development ecosystem for building and deploying the Air Quality Prediction System. Python provided analytical strength, Google Colab offered scalable cloud-based execution, machine learning libraries enabled intelligent modeling, and Gradio facilitated end-user interaction. Together, these technologies ensured that the system is efficient, scalable, reproducible, and practically deployable.

DESCRIPTIVE ANALYSIS

7.Descriptive Analysis

Descriptive analysis was performed to understand the overall characteristics and distribution of the air quality dataset before applying machine learning models. This step helped in identifying trends, variations, correlations, and potential outliers within the environmental parameters. Both statistical measures and graphical visualizations were used to obtain meaningful insights from the data.

Statistical Summary

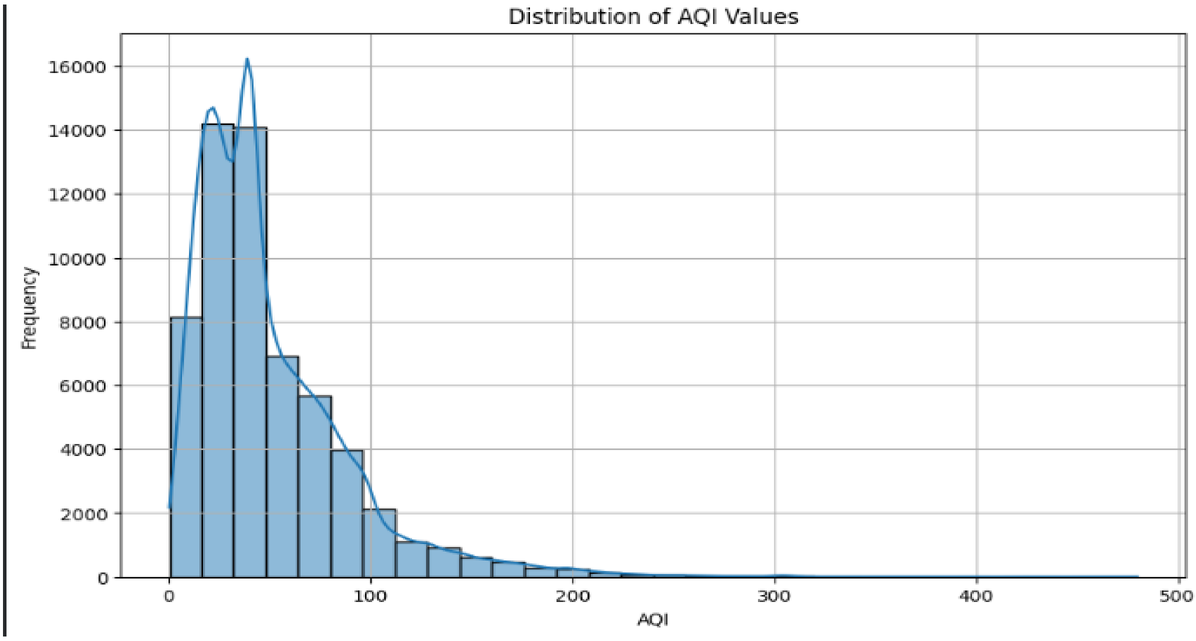
A statistical summary of the dataset was generated to analyse the central tendency and dispersion of the features. Measures such as mean, median, minimum, maximum, and standard deviation were calculated for all numerical attributes. The mean and median values helped in understanding the average pollution levels, while the minimum and maximum values indicated the range of variation. The standard deviation provided insights into the variability of pollutant concentrations. Higher standard deviation values indicated large fluctuations in pollution levels, whereas lower values represented stable environmental conditions. This statistical overview helped in identifying features with high variability and detecting potential irregularities in the dataset.

```
Statistics Summary
        PM2.5 (ug/m3)  PM10 (ug/m3)    NO (ug/m3)   NO2 (ug/m3)     NOx (ppb)  \
count    54323.000000  54450.000000  55153.000000  55100.000000  55315.000000
mean        29.718423     58.531252     13.077963     39.195409     31.171453
std         21.365617     32.388101     16.124784     30.716953     25.048555
min          0.250000      1.000000      0.100000      0.100000      0.000000
25%         13.250000     34.000000      3.800000     17.400000     13.780000
50%         24.000000     53.500000      8.100000     30.650000     24.880000
75%         42.500000     78.500000     16.170000     52.950000     41.150000
max        449.500000    929.000000    288.170000    313.650000    364.800000

        NH3 (ug/m3)   SO2 (ug/m3)    CO (mg/m3)  Ozone (ug/m3)  \
count  53564.000000  54285.000000  54673.000000   54567.000000
mean      10.057892      5.782241      0.634884      28.681448
std        5.949501      4.538823      0.850052      19.189147
min        0.100000      0.030000      0.000000       0.070000
25%        6.200000      3.650000      0.350000      16.370000
50%        8.420000      5.150000      0.570000      22.380000
75%       12.200000      6.970000      0.790000      36.000000
max      195.200000    199.770000     41.100000     199.700000
```

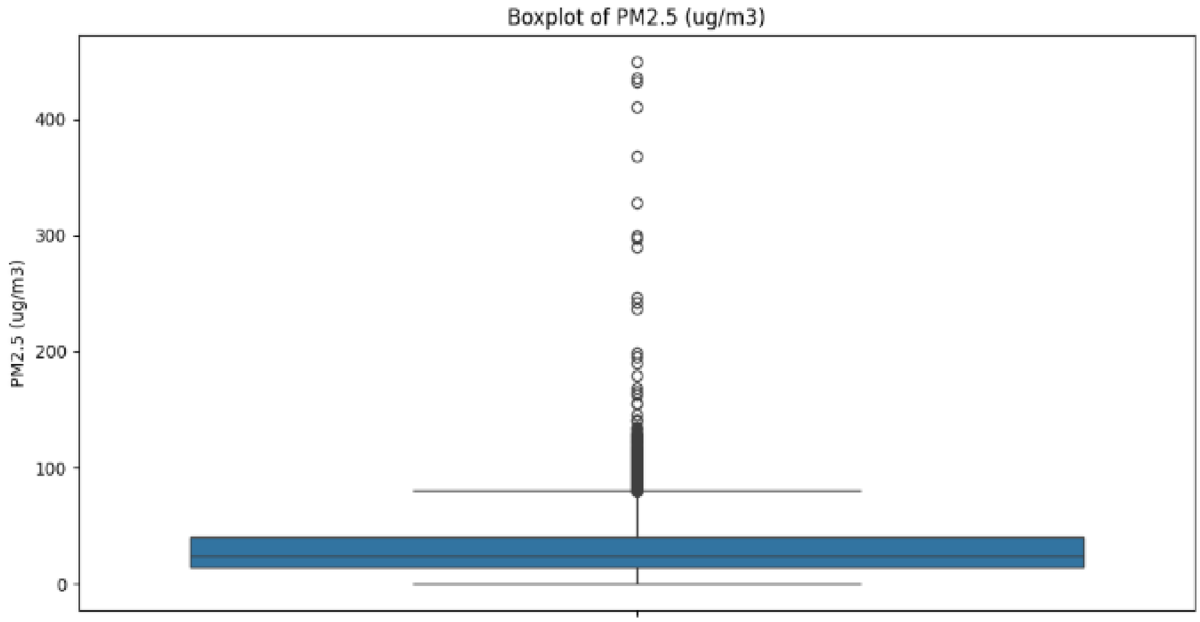
Distribution Analysis (Histogram)

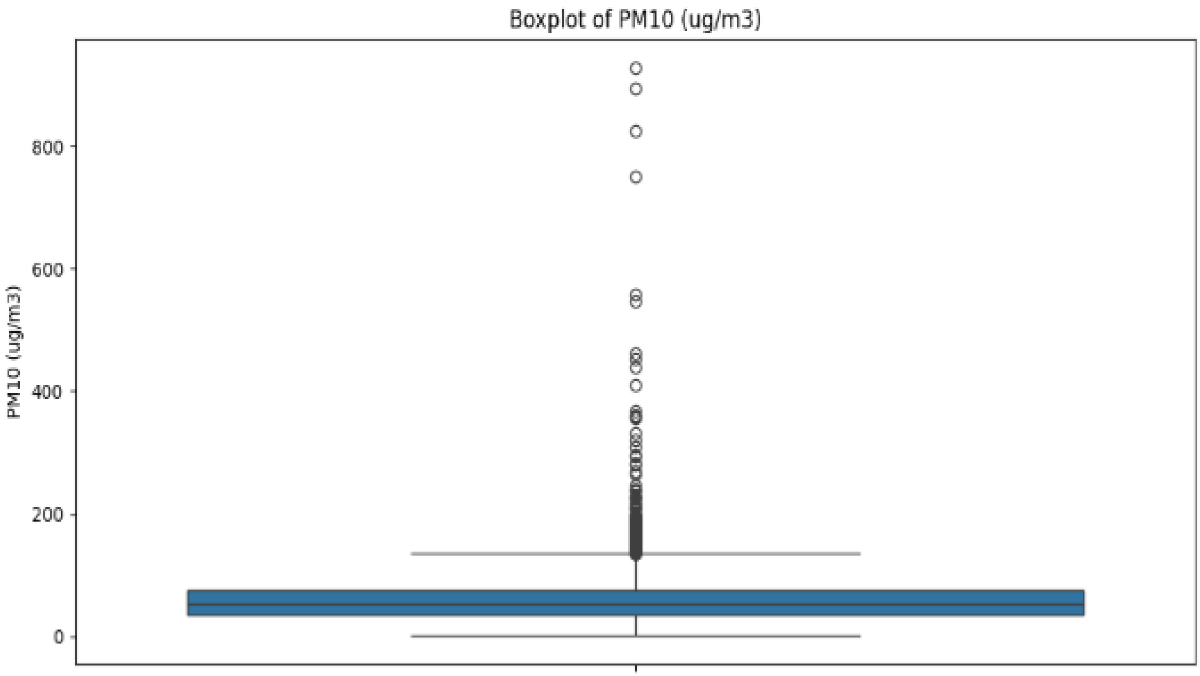
Histograms were plotted for major attributes such as AQI and particulate matter concentrations to examine their distribution patterns. The histograms revealed whether the data followed a normal distribution or was skewed. In environmental datasets, pollutant levels are often right-skewed due to occasional extreme pollution events. Understanding the distribution pattern was important for selecting appropriate preprocessing techniques and machine learning algorithms. This analysis also helped in identifying concentration ranges where most observations were clustered.

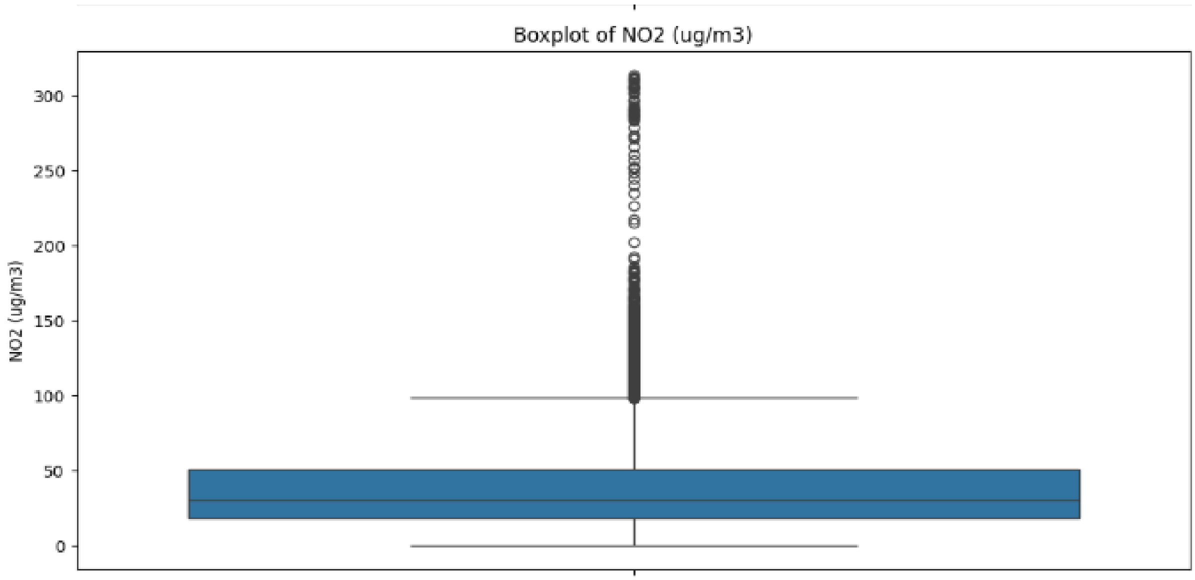


Outlier Detection (Boxplot Analysis)

Boxplots were generated to detect the presence of outliers in key pollutant parameters. Environmental data often contains extreme values due to sudden pollution spikes, industrial emissions, or abnormal weather conditions. The boxplot visualization highlighted the interquartile range and extreme values beyond the whiskers. Identifying outliers was essential to determine whether they represented genuine environmental events or measurement errors. This step supported better decision-making during the preprocessing phase.

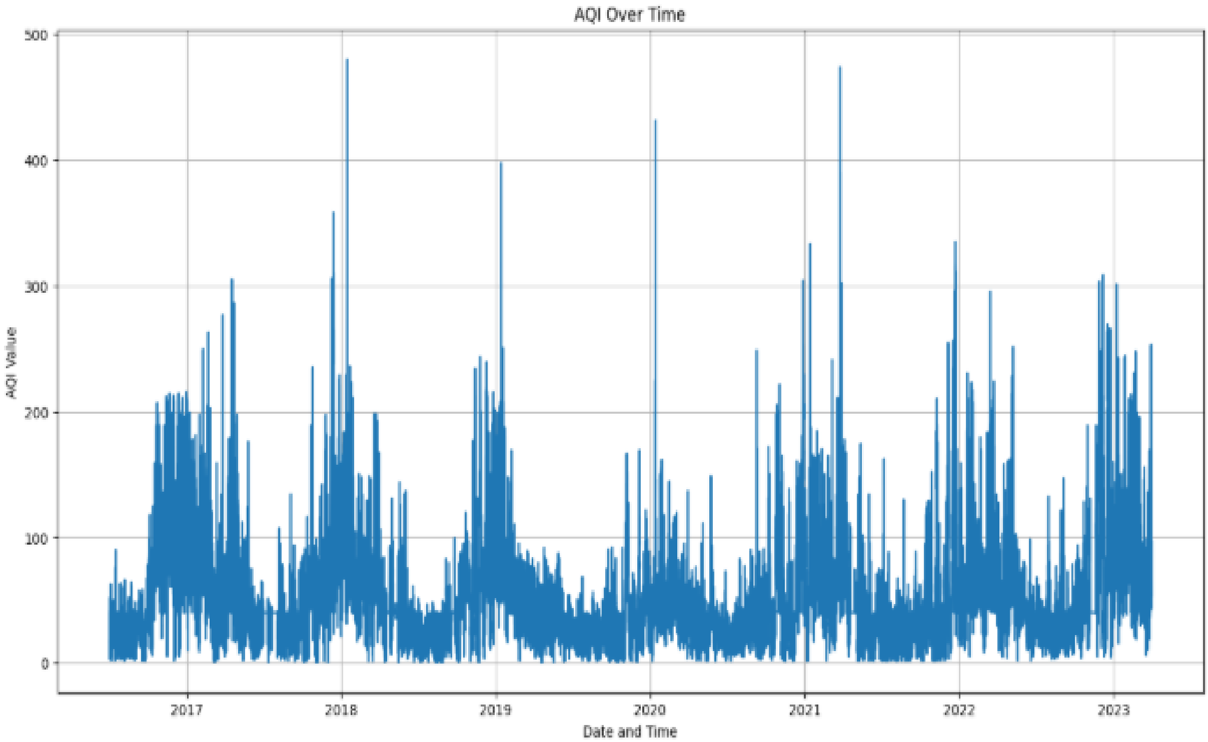

Boxplot of PM2.5 (ug/m3)
PM2.5 (ug/m3)
400
300
200
100
0


Boxplot of PM10 (ug/m3)
PM10 (ug/m3)
800
600
400
200
0



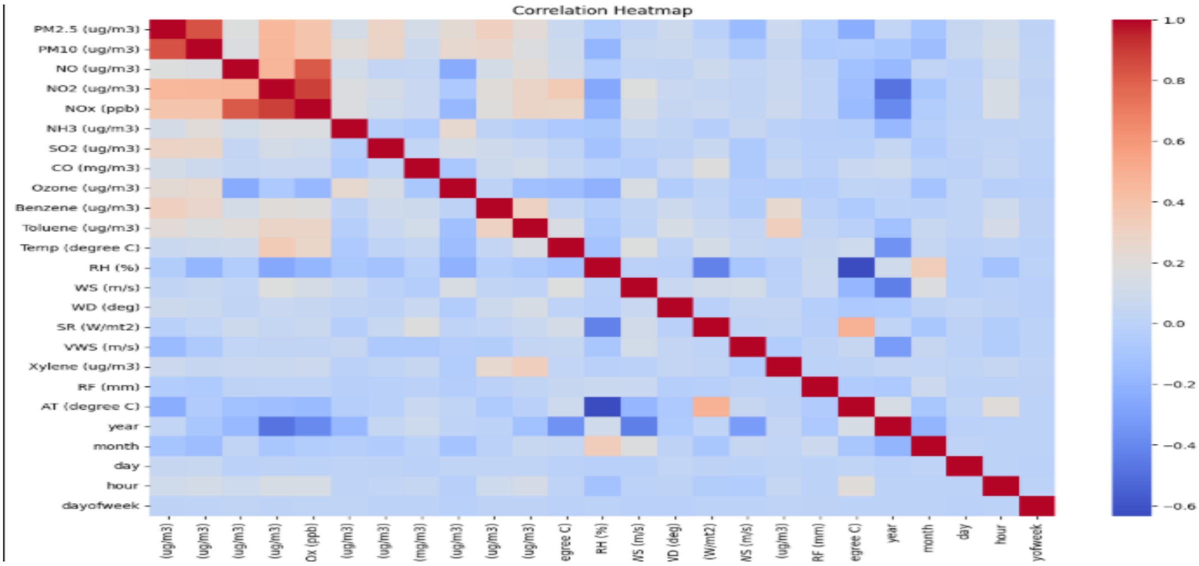
Trend Analysis (Line Plot Over Time)

Since the dataset contained date-related information, time-series line plots were created to analyse pollution trends over time. AQI values were plotted against date to observe seasonal variations and long-term patterns. The trend analysis helped in identifying periods of high pollution, possible seasonal fluctuations, and gradual increases or decreases in air quality levels. Such insights are valuable for understanding environmental behaviour and improving predictive model performance.



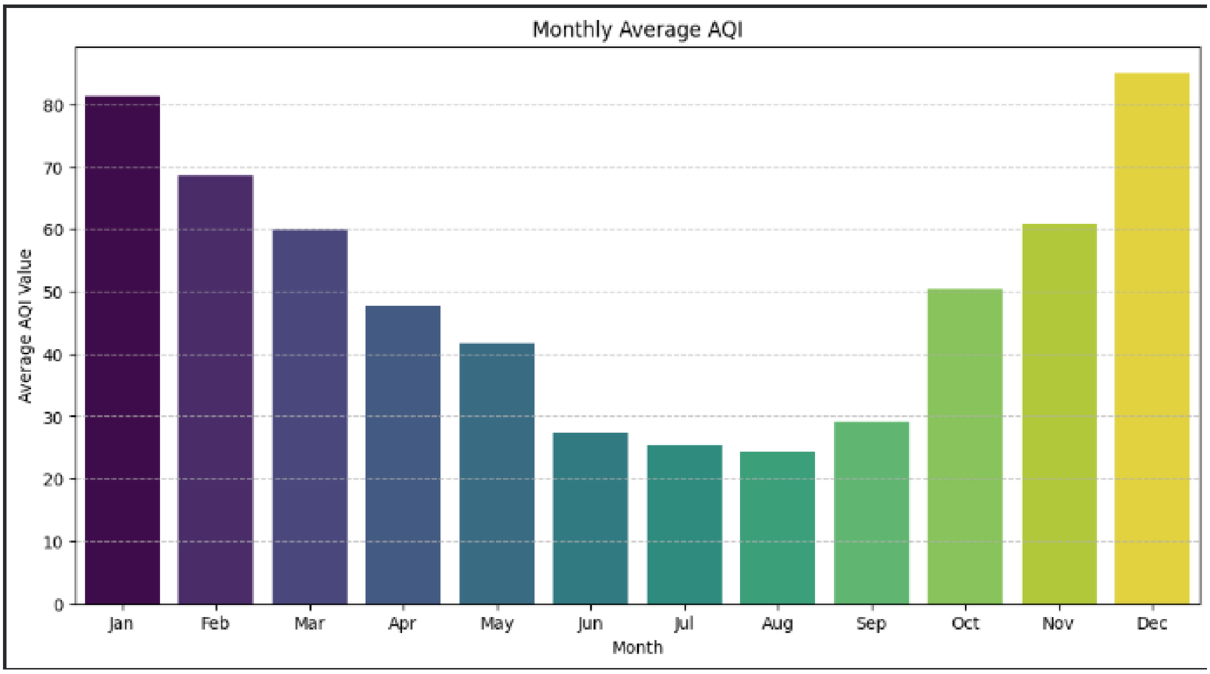
Correlation Analysis (Heatmap)

A correlation heatmap was generated to examine the relationships between different environmental parameters. This visualization helped in identifying strongly correlated variables and understanding how individual pollutants influence the AQI. Strong positive correlations were observed between particulate matter concentrations and AQI levels, indicating that these pollutants significantly impact overall air quality. The heatmap also helped in detecting multicollinearity among independent variables, which is important before training machine learning models.



Monthly Average AQI Analysis (Bar Chart)

The above bar chart represents the average Air Quality Index (AQI) values recorded across different months. This visualization was created to analyse seasonal variations and identify patterns in air pollution levels throughout the year. From the graph, it can be observed that AQI levels are comparatively higher during the beginning and end of the year, particularly in January, November, and December. Among all months, December shows the highest average AQI value, indicating poorer air quality conditions during this period. This increase may be attributed to seasonal factors such as lower temperatures, reduced wind speed, and increased human activities. In contrast, the mid-year months such as June, July, and August show significantly lower AQI values. August records the lowest average AQI, suggesting relatively better air quality conditions during this period. This reduction may be associated with increased rainfall and improved atmospheric dispersion. Overall, the bar chart clearly highlights seasonal fluctuations in air pollution levels. Understanding these monthly variations is important for identifying high-risk periods and improving the predictive capability of the air quality forecasting model.



SYSTEM DESIGN

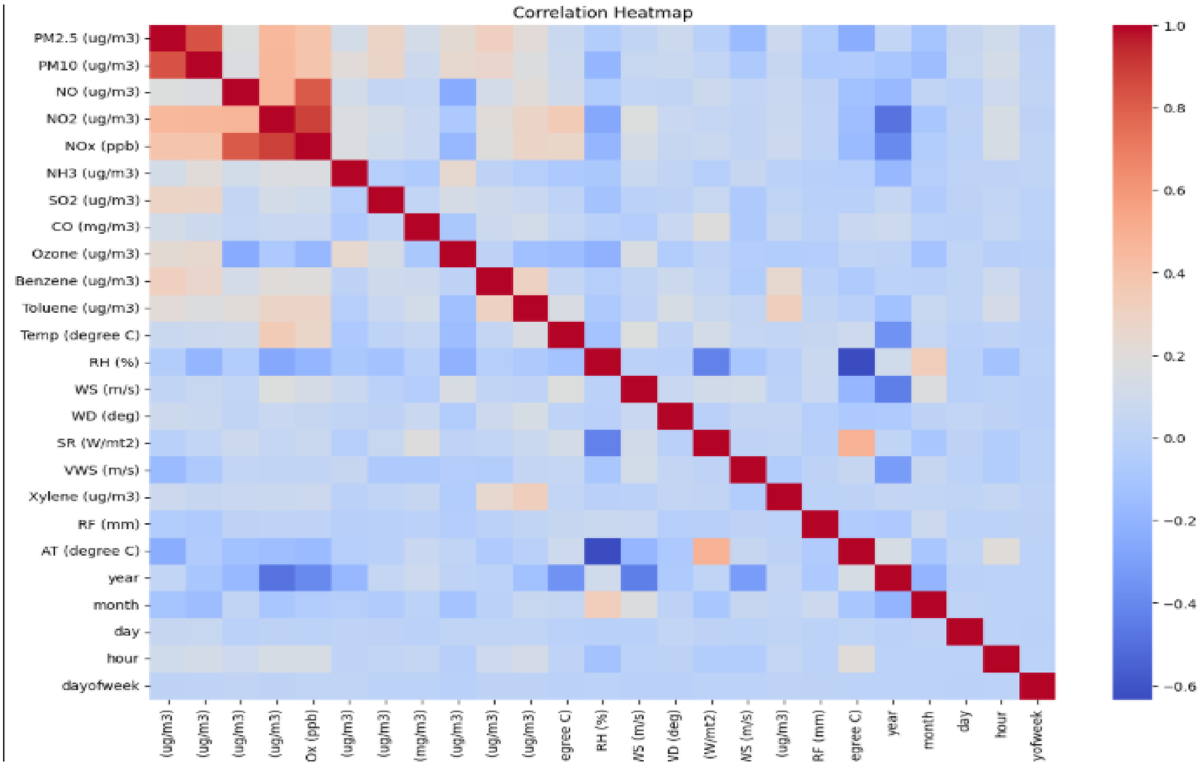
8.1 Identify Relationship Between Variables

The Identify Relationship Between Variables module is designed to analyse and understand the associations between different pollutants, meteorological parameters, and the Air Quality Index (AQI). Since air pollution is influenced by multiple interacting factors, it is essential to examine how each variable contributes to overall air quality levels. This analysis helps in determining which pollutants have the strongest impact on AQI and how environmental conditions influence pollution concentration. To achieve this, statistical and exploratory data analysis techniques such as correlation analysis are applied. Correlation measures the strength and direction of the relationship between two variables, indicating whether they move together positively, negatively, or independently. A positive correlation between a pollutant (such as PM2.5 or PM10) and AQI suggests that as the pollutant concentration increases, the AQI value also increases, indicating worsening air quality. Conversely, a negative correlation may indicate that certain meteorological factors, such as wind speed, help reduce pollutant concentration and thereby lower AQI levels. Understanding these relationships is crucial for feature selection in machine learning models. Variables that show strong correlation with AQI are considered significant predictors and are retained for model training, while weakly related or redundant features may be removed to improve model efficiency. Additionally, analysing interrelationships among pollutants helps in identifying multicollinearity issues, where highly correlated independent variables may affect model performance. Overall, identifying the relationships between variables provides valuable insights into pollution dynamics, supports data-driven decision making, and enhances the predictive accuracy of the AQI forecasting model. This step strengthens the foundation of the project by ensuring that the model is built on meaningful and scientifically relevant features.

Correlation Analysis (Heatmap)

A correlation heatmap was generated to examine the relationships between different environmental parameters. This visualization helped in identifying strongly correlated variables and understanding how individual pollutants influence the AQI.

Strong positive correlations were observed between particulate matter concentrations and AQI levels, indicating that these pollutants significantly impact overall air quality. The heatmap also helped in detecting multicollinearity among independent variables, which is important before training machine learning models.



Relationship Between Pollutants and AQI

To understand how different air pollutants influence the Air Quality Index (AQI), a correlation and scatter plot analysis was performed between major pollutants and AQI values. The pollutants considered in this study include PM2.5, PM10, NO_2, SO_2, CO, and Ozone (O_3). Scatter plots were generated to visually analyse the relationship between each pollutant concentration and AQI levels.

PM2.5 vs AQI

The scatter plot shows a strong positive linear relationship between PM2.5 concentration and AQI. As PM2.5 levels increase, AQI values increase significantly. PM2.5 contributes heavily to AQI because:

- It consists of fine particulate matter that can penetrate deep into the lungs.
- Even small increases in PM2.5 concentration result in a noticeable rise in AQI.

- High AQI values (above 300) are strongly associated with high PM2.5 levels.

This indicates that PM2.5 is one of the most influential pollutants in determining air quality.

PM10 vs AQI

PM10 also shows a positive correlation with AQI, but the relationship is slightly more scattered compared to PM2.5.

- Higher PM10 concentrations generally correspond to higher AQI values.
- However, variability suggests that PM10 alone does not determine AQI as strongly as PM2.5.

PM10 contributes to respiratory problems but has a slightly lower impact on AQI compared to finer particles.

NO_2 vs AQI

NO_2 shows a moderate relationship with AQI

- AQI increases in some regions with increasing NO_2 levels.
- However, the data distribution is more scattered, indicating that NO_2 contributes to AQI but is not the dominant factor.

NO_2 mainly originates from vehicle emissions and industrial combustion processes.

SO_2 vs AQI

SO_2 shows a weak to moderate correlation with AQI.

- At lower SO_2 levels, AQI varies widely.
- Some high AQI values are observed even when SO_2 levels are moderate, suggesting other pollutants influence AQI more strongly.

SO_2 mainly comes from fossil fuel burning in power plants and industries.

CO vs AQI

CO demonstrates a weak correlation with AQI.

- Most AQI values remain moderate despite varying CO concentrations.
- Only at higher CO concentrations does AQI show noticeable increase.

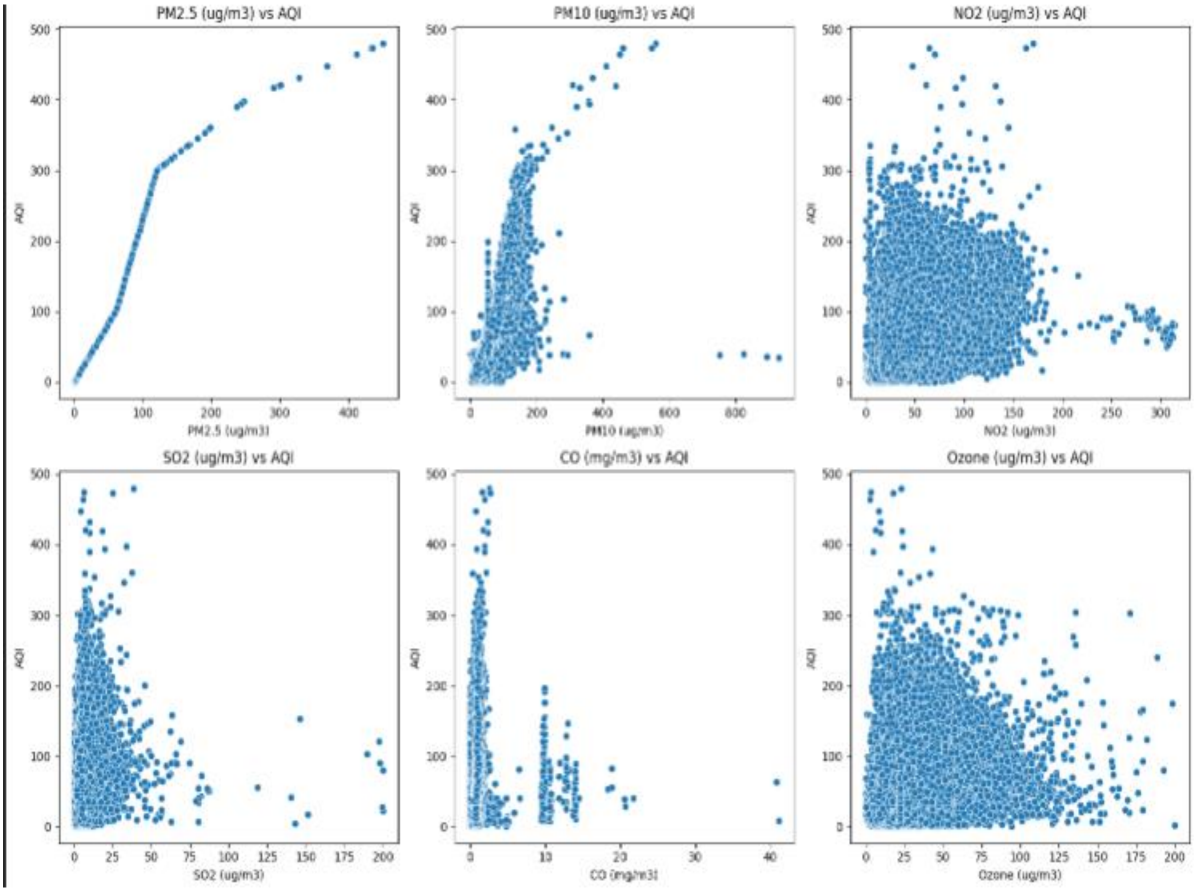
This indicates CO contributes to AQI but is not a primary driver in most cases.

Ozone (O_3) vs AQI

Ozone shows a moderate and slightly negative/complex relationship in certain ranges.

- AQI increases in some cases with rising Ozone levels.
- However, the relationship is not strictly linear.

Ozone impact may depend on temperature and sunlight conditions.



Multicollinearity Analysis

Multicollinearity analysis was performed to identify highly correlated independent variables in the Air Quality dataset. Since multiple pollutants and meteorological variables are included, there is a possibility that some features carry overlapping information.

To detect multicollinearity, the Variance Inflation Factor (VIF) method was used.

VIF Formula: VIF= 1/ 1-R^2

Where R^2 represents the coefficient of determination obtained by regressing one independent variable against all other independent variables.

VIF Interpretation Criteria

- VIF = 1 → No multicollinearity
- VIF between 1–5 → Low correlation
- VIF between 5–10 → Moderate correlation
- VIF > 10 → High multicollinearity (Problematic)

Observed VIF Results

The VIF analysis produced the following key observations

Extremely High VIF (Severe Multicollinearity)

- Year → 555.90
- NOx (ppb) → 470.07
- NO_2 (µg/m³) → 212.35
- AT (°C) → 188.48
- Temperature (°C) → 152.33
- Relative Humidity (RH) → 89.57
- NO (µg/m³) → 81.45

These values are far above the acceptable threshold (10), indicating severe multicollinearity.

Moderate to High VIF

- PM10 → 17.69
- PM2.5 → 13.19

These also indicate strong correlation but are comparatively lower than NOx-related features.

Acceptable VIF (Below 10)

Features such as:

- Ozone
- Wind Speed
- Wind Direction
- SO_2
- CO
- Benzene
- Xylene
- Toluene
- Hour, Day, Month

index	feature	VIF
20	year	555.9042343408053
4	NOx (ppb)	470.0702945766892
3	NO2 (ug/m3)	212.35743548951368
19	AT (degree C)	188.48874088542365
11	Temp (degree C)	152.33504725889324
12	RH (%)	89.57197967752982
2	NO (ug/m3)	81.45204617042523
1	PM10 (ug/m3)	17.697574110884
0	PM2.5 (ug/m3)	13.195619055747098
14	WD (deg)	7.611102894754166
21	month	5.733615011105244
8	Ozone (ug/m3)	5.062424067260679
13	WS (m/s)	5.036294720165858
5	NH3 (ug/m3)	4.746612475455947
23	hour	4.413264757415237
22	day	4.229683068404574
24	dayofweek	3.2585553030608536
6	SO2 (ug/m3)	3.0779617221772027
10	Toluene (ug/m3)	2.2161668652176862
15	SR (W/mt2)	2.05651461657696
7	CO (mg/m3)	1.7403174800538974
9	Benzene (ug/m3)	1.6522166175081368
17	Xylene (ug/m3)	1.19906341469294
16	VWS (m/s)	1.1196836428986847
18	RF (mm)	1.0363816824646872

Impact on Feature Selection

Feature selection is a crucial step in building a reliable and efficient machine learning model. After performing Multicollinearity Analysis (VIF) and Feature Importance Analysis, the most relevant variables influencing AQI were identified. The feature importance graph clearly shows that some features contribute significantly to AQI prediction, while many others have very minimal impact.

Observations from Feature Importance Analysis

From the generated feature importance plot

1. PM10 (µg/m³) – Highest Influence

- PM10 has the highest importance score (approximately 0.72).
- This indicates that PM10 is the most dominant feature in predicting AQI in the trained model.
- It contributes the majority of the predictive power.

2. Secondary Influential Features

Features such as:

- Benzene
- AT (Temperature)
- Month
- VWS (Wind Speed)
- CO

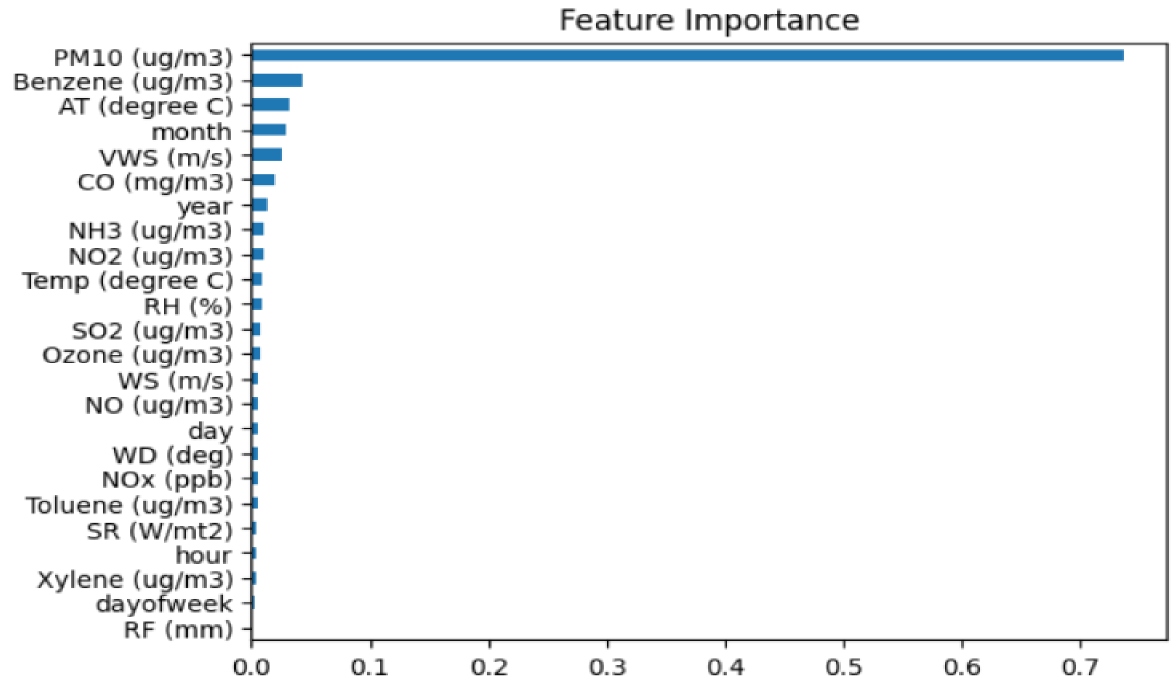
have relatively smaller but noticeable contributions compared to PM10.

3. Low Impact Features

Most of the remaining features such as:

- NOx
- Toluene
- SR (Solar Radiation)
- Hour
- Xylene
- Day of week
- Rainfall

have extremely low importance scores.


Feature Importance
PM10 (ug/m3)
Benzene (ug/m3)
AT (degree C)
month
VWS (m/s)
CO (mg/m3)
year
NH3 (ug/m3)
NO2 (ug/m3)
Temp (degree C)
RH (%)
SO2 (ug/m3)
Ozone (ug/m3)
WS (m/s)
NO (ug/m3)
day
WD (deg)
NOx (ppb)
Toluene (ug/m3)
SR (W/mt2)
hour
Xylene (ug/m3)
dayofweek
RF (mm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7

DATA IMPLEMENTATION

9.Data Implementation

9.1 Source Code

AQI Calculation Using PM2.5

The AQI Calculation Using PM2.5 module is responsible for computing the Air Quality Index based specifically on the concentration of fine particulate matter (PM2.5) present in the dataset. PM2.5 refers to particulate matter with a diameter of less than 2.5 micrometres, which can penetrate deep into the lungs and pose serious health risks. Since PM2.5 is one of the most critical pollutants influencing air quality, it is widely used as a primary indicator in AQI computation. In this project, the AQI value is derived by applying the standard breakpoint-based linear interpolation method, which maps pollutant concentration ranges to corresponding AQI index ranges. The calculation process involves identifying the concentration interval in which a given PM2.5 value falls and then applying a linear scaling formula to convert that concentration into an AQI value. Each concentration range corresponds to a specific AQI category, such as Good, Satisfactory, Moderate, Poor, Very Poor, or Severe. The interpolation formula ensures a smooth transition between AQI ranges rather than assigning fixed values, thereby providing a more accurate and continuous index representation. This approach preserves the proportional relationship between pollutant concentration and air quality severity. Once the AQI value is calculated for each PM2.5 observation, it is stored as a new feature in the dataset. This computed AQI serves as a quantitative representation of air pollution levels and forms the basis for further processes such as AQI categorization, visualization, and machine learning-based prediction. By implementing this method, the system ensures that AQI values are systematically derived from scientifically defined standards, improving the reliability and interpretability of the air quality assessment.

A custom function calculate_pm25_aqi() was implemented to

- Identify the PM2.5 concentration range
- Apply linear interpolation formula
- Return corresponding AQI value

```python
def calculate_pm25_aqi(pm25):
    if pm25 <= 30:
        return (50/30) * pm25
    elif pm25 <= 60:
        return ((100-51)/(60-31))*(pm25-31) + 51
    elif pm25 <= 90:
        return ((200-101)/(90-61))*(pm25-61) + 101
    elif pm25 <= 120:
        return ((300-201)/(120-91))*(pm25-91) + 201
    elif pm25 <= 250:
        return ((400-301)/(250-121))*(pm25-121) + 301
    else:
        return ((500-401)/(500-251))*(pm25-251) + 401
aqi['AQI'] = aqi['PM2.5 (ug/m3)'].apply(calculate_pm25_aqi)
```

- The function checks which AQI breakpoint range the PM2.5 value belongs to.
- Linear interpolation is applied.
- The calculated AQI value is stored in a new column called AQI.

AQI Category Classification

To convert numerical AQI value into meaningful air quality categories.

AQI Range	Category
0–50	Good
51–100	Satisfactory
101–200	Moderate
201–300	Poor
301–400	Very Poor
401–500	Severe

```python
#AQI Category
def aqi_category(aqi):
    if aqi <= 50:
        return 'Good'
    elif aqi <= 100:
        return 'Satisfactory'
    elif aqi <= 200:
        return 'Moderate'
    elif aqi <= 300:
        return 'Poor'
    elif aqi <= 400:
        return 'Very Poor'
    else:
        return 'Severe'

aqi['AQI_Category'] = aqi['AQI'].apply(aqi_category)
aqi.head()
```

- The function maps AQI value into predefined pollution categories.
- The category is stored in a new column AQI_Category.
- This improves interpretability for users.

Current AQI value prediction

The Current AQI Value Prediction module is designed to estimate the Air Quality Index (AQI) using pollutant concentration levels and environmental parameters available in the dataset. Since AQI is a continuous numerical value that reflects the overall air pollution status, a supervised machine learning regression approach is adopted. In this project, the Random Forest Regressor algorithm is used to model the relationship between multiple air pollutants such as PM2.5, PM10, NO_2, SO_2, CO, Ozone, and other meteorological features like temperature and wind speed, and the corresponding AQI value. The dataset is first divided into independent variables (features) and a dependent variable (target). The AQI column is selected as the target variable, while all other pollutant and environmental parameters are used as predictors. To ensure proper model evaluation and to avoid overfitting, the dataset is split into training and testing subsets using an 80:20 ratio. The training data is used to train the Random Forest model, and the testing data is used to evaluate its predictive performance. Random Forest Regressor is an ensemble learning technique that constructs multiple decision trees during training and combines their outputs to produce a more accurate and stable prediction. Each tree learns different patterns from subsets of the data, and the final AQI prediction is obtained by averaging the predictions of all trees. This approach reduces variance, improves generalization, and handles nonlinear relationships effectively, which is particularly important in air quality datasets where pollutant interactions are complex and not strictly linear. By using 300 decision trees, the model enhances prediction robustness and minimizes overfitting. Once trained, the model predicts AQI values for unseen data, thereby estimating the current air quality conditions based on pollutant levels. The predicted AQI values can further be categorized into air quality levels such as Good, Moderate, or Poor for better interpretation. Overall, this module successfully captures the complex relationship between pollutant concentrations and AQI, providing an accurate and reliable estimation of current air quality conditions using machine learning techniques.

Feature Selection

The dataset was divided into:

- Independent Variables (X) → All pollutant and environmental features
- Dependent Variable (y) → AQI value

```
X = aqi.drop(columns=['AQI', 'AQI_Category'])
y = aqi['AQI']
```

- AQI is the target variable.
- AQI_Category is removed to avoid data leakage.
- Remaining features like PM2.5, PM10, NO2, CO, SO2, Temperature, etc., are used as predictors.

Train-Test Splitting

To evaluate the model performance, the dataset is divided into training and testing sets.

```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

- 80% data → Training
- 20% data → Testing
- random_state = 42 ensures reproducibility

Model Selection – Random Forest Regressor

Random Forest Regressor is an ensemble learning algorithm that:

- Builds multiple decision trees
- Combines their outputs
- Reduces overfitting
- Improves prediction accuracy

```python
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    n_estimators=200,
    random_state=42,
    class_weight='balanced'
)

rf_model.fit(X_train_smote, y_train_smote)
```

```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', n_estimators=200,
                       random_state=42)
```

AQI Prediction

After training, the model predicts AQI values on test data

```python
y_pred = rf_model.predict(X_test)
```

AQI for Next 24 Hours Prediction

The AQI for Next 24 Hours Prediction module is developed to forecast future air quality conditions based on historical pollutant concentrations and environmental parameters. Unlike the current AQI calculation, which reflects real-time pollution levels, this module aims to estimate the AQI value for the upcoming 24 hours using machine learning techniques. Predicting future AQI is crucial for proactive environmental monitoring, public health planning, and early warning systems, as it allows individuals and authorities to take preventive measures in advance. To implement this forecasting mechanism, the dataset is structured in such a way that historical pollutant readings and meteorological variables serve as input features, while the AQI value of the next time period is treated as the target variable. By shifting the AQI column forward by 24 hours, the model learns the temporal relationship between current pollutant levels and future air quality conditions. This transformation enables the algorithm to capture patterns such as pollutant accumulation, dispersion influenced by wind speed, and temperature variations that significantly impact air pollution trends. A Random Forest-based regression model is employed to perform the prediction because of its ability to model complex nonlinear relationships among multiple variables. Air pollution dynamics are influenced by interactions between different pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and meteorological conditions, which are rarely linear. The ensemble nature of the Random Forest algorithm allows it to aggregate multiple decision trees, thereby improving prediction stability and reducing overfitting. Once trained, the model generates a predicted AQI value representing the expected air quality after 24 hours. The predicted numerical AQI value is further classified into standard air quality categories such as Good, Moderate, or Severe to enhance interpretability. This forward-looking prediction capability strengthens the overall system by not only assessing present air quality but also providing insights into future pollution levels. As a result, the Next 24 Hours AQI Prediction module transforms the project from a simple monitoring system into a predictive decision-support tool for environmental management.

Data Preprocessing Module

- Data is sorted based on year, month, day, and hour to maintain time sequence.
- The target variable (AQI_next_24h) is created using time shifting.
- The last 24 rows are removed because they do not have future AQI values.
- This prepares the dataset for time-series-based prediction.

```
# Sort by time features instead of datetime
df = aqi.sort_values(by=['year', 'month', 'day', 'hour'])
```

```
# Shift AQI by -24 to get next 24-hour AQI
df['AQI_next_24h'] = aqi['AQI'].shift(-24)

# Drop last 24 rows (they won't have target)
df = df.dropna()
```

Feature Selection Module

- Important pollutant and meteorological features are selected.
- X contains independent variables.
- Y contains the target variable (Next 24-hour AQI).
- Feature selection was based on correlation and feature importance analysis.

```
feature_cols = [
    'PM2.5 (ug/m3)', 'PM10 (ug/m3)', 'NO2 (ug/m3)', 'SO2 (ug/m3)',
    'CO (mg/m3)', 'Ozone (ug/m3)', 'Temp (degree C)', 'RH (%)',
    'WS (m/s)', 'hour', 'dayofweek'
]
X = df[feature_cols]
y = df['AQI_next_24h']
```

Data Splitting Module

Dataset is split into:

- 80% training data
- 20% testing data

random_state ensures reproducibility.

```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Model Training Module

- Random Forest Regressor is used because AQI is a continuous value.
- n_estimators=300 means 300 decision trees are created.
- The model learns the relationship between pollutants and future AQI.
- The trained model predicts next 24-hour AQI value.

```python
from sklearn.ensemble import RandomForestRegressor

rf_24h = RandomForestRegressor(
    n_estimators=300,
    random_state=42
)

rf_24h.fit(X_train, y_train)
```

```
RandomForestRegressor
RandomForestRegressor(n_estimators=300, random_state=42)
```

AQI Category Classification Module

- After predicting AQI value, category is determined.
- Based on category, health advisory message is generated.
- This improves interpretability for end users.

```python
import numpy as np

# Define the health_alert function based on AQI categories
def health_alert(category):
    if category == 'Good':
        return 'Minimal impact'
    elif category == 'Satisfactory':
        return 'Minor breathing discomfort to sensitive people'
    elif category == 'Moderate':
        return 'Breathing discomfort to people with lung disease, heart disease, children and older adults; impacts may be experienced by healthy peopl
    elif category == 'Poor':
        return 'Breathing discomfort to most people on prolonged exposure; may cause breathing discomfort to people with heart disease'
    elif category == 'Very Poor':
        return 'Respiratory illness on prolonged exposure; impacts may be experienced by healthy people also'
    else: # Severe
        return 'Respiratory effects likely to affect healthy people and seriously impact those with existing diseases'
```

Prediction Module

This generates Predicted AQI value for next 24 hours. Then Convert AQI → Category Generate health alert.

```python
user_input = np.array([[85.0, 140.0, 32.0, 18.0, 40.0, 15.0, 18.0, 0.9, 25.0, 3.0, 5.0]])

predicted_aqi_24h = rf_24h.predict(user_input)[0]

# Convert to category and get alert
# Use the already defined aqi_category function from cell 6DQtcoechpeZ
predicted_category_24h = aqi_category(predicted_aqi_24h)
alert_message_24h = health_alert(predicted_category_24h)

print("Predicted AQI after 24 hours:", round(predicted_aqi_24h,2))
print("AQI Category after 24 hours:", predicted_category_24h)
print("Health Alert after 24 hours:", alert_message_24h)
```

9.2 Data Visualization and Analytical Insights

The Data Visualization and Analytical Insights module plays a crucial role in understanding the structure, distribution, and interaction of variables within the air quality dataset. Visualization techniques were employed to transform raw numerical data into meaningful graphical representations, enabling clearer interpretation of pollutant behaviour, spatial distribution, and inter-variable relationships. In this project, advanced visualization methods such as Pairwise Relationship Analysis (Pair plot), Wind Rose Plot, and Geographical Pollution Map were used to derive analytical insights. The Pairwise Relationship Analysis (Pair plot) was utilized to examine the relationships between key pollutants and AQI simultaneously. This visualization provides both distribution plots and scatter plots for multiple variables in a single matrix format. Through this analysis, strong positive relationships were observed between particulate matter (PM2.5 and PM10) and AQI, confirming their dominant influence on air quality. The pair plot also helped in identifying potential multicollinearity between pollutant variables, as well as detecting patterns, clusters, and outliers in the dataset. This step supported informed feature selection and strengthened the machine learning modeling process. The Wind Rose Plot was implemented to analyse the relationship between wind direction, wind speed, and pollutant dispersion. Since meteorological factors significantly influence pollution concentration, understanding wind behaviour is essential in air quality studies. The wind rose visually represents how pollutant levels vary with different wind directions and intensities. From this visualization, it becomes easier to identify prevailing wind patterns and determine whether certain directions contribute to higher pollution levels. This insight helps in understanding pollutant transport mechanisms and environmental dynamics. The Geographical Pollution Map was used to represent spatial variations in air quality across different locations. By plotting AQI or pollutant concentrations on a map, spatial hotspots and low-pollution regions were identified. This visualization highlights geographical patterns and allows comparison between different monitoring areas. It enhances interpretability by showing how pollution levels vary across regions rather than just over time. Such spatial analysis is particularly useful for environmental planning and policy decision-making. Overall, these visualization techniques provided comprehensive analytical insights into pollutant behaviour, environmental influences, and spatial distribution patterns. They strengthened the exploratory data analysis process, supported feature selection, and improved the overall interpretability of the Air Quality Prediction System.

Pairwise Relationship Analysis (Pair plot)

To understand the relationships between major air pollutants and AQI, a pair plot visualization was generated. The pair plot provides a comprehensive view of

- Individual feature distributions (diagonal plots)
- Pairwise relationships between variables (scatter plots)
- Correlation patterns between pollutants and AQI

The key variables included in this analysis were:

- PM2.5
- PM10
- NO_2
- CO
- AQI

Observations from Pair plot

PM2.5 and AQI

- A strong positive linear relationship is observed.
- As PM2.5 concentration increases, AQI increases significantly.
- The points form an upward trend, indicating high correlation.

PM10 and AQI

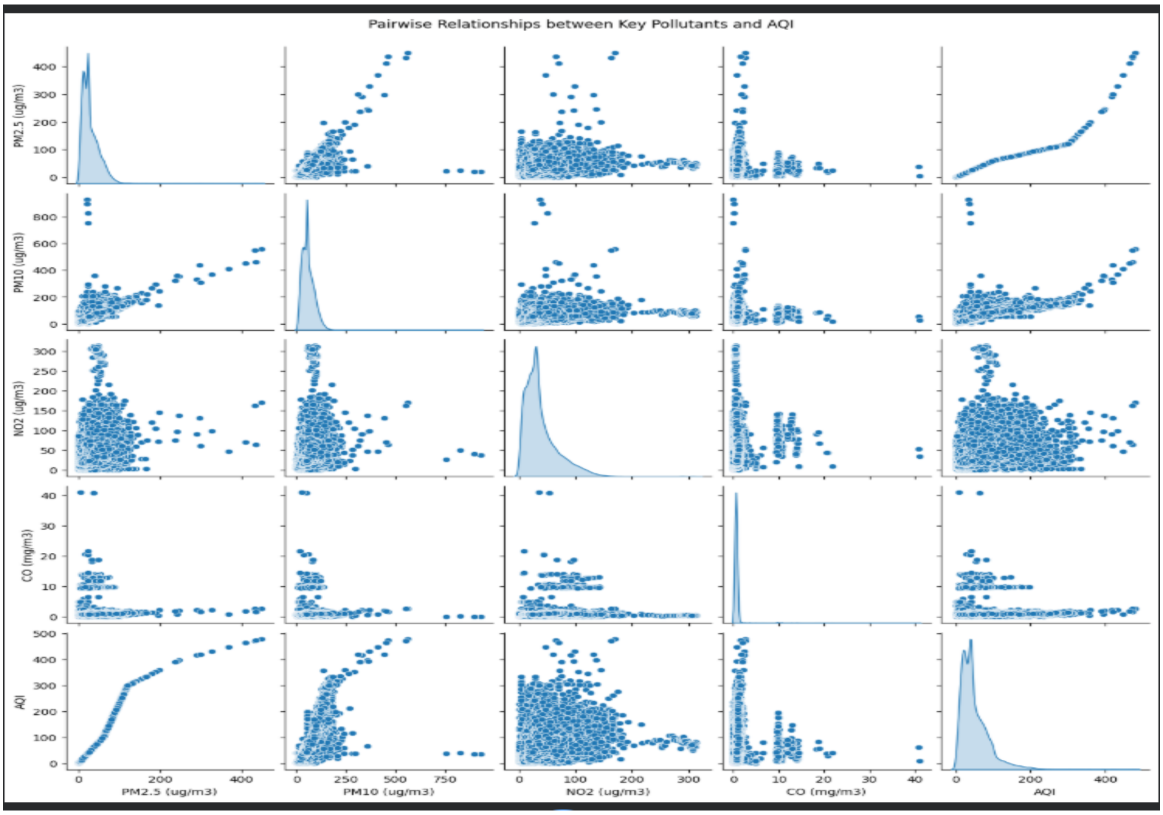
- A clear positive relationship is visible.
- AQI increases as PM10 concentration increases.
- The relationship is slightly more scattered compared to PM2.5.

NO_2 and AQI

- The relationship appears moderate and more scattered.
- AQI increases in certain ranges of NO_2, but not strictly linearly.
- Indicates NO_2 contributes to AQI but is not the dominant factor.

CO AQI

- The relationship between CO and AQI is weak.
- Most **and** AQI values remain moderate even when CO varies.
- Only extreme CO values show noticeable AQI increase.



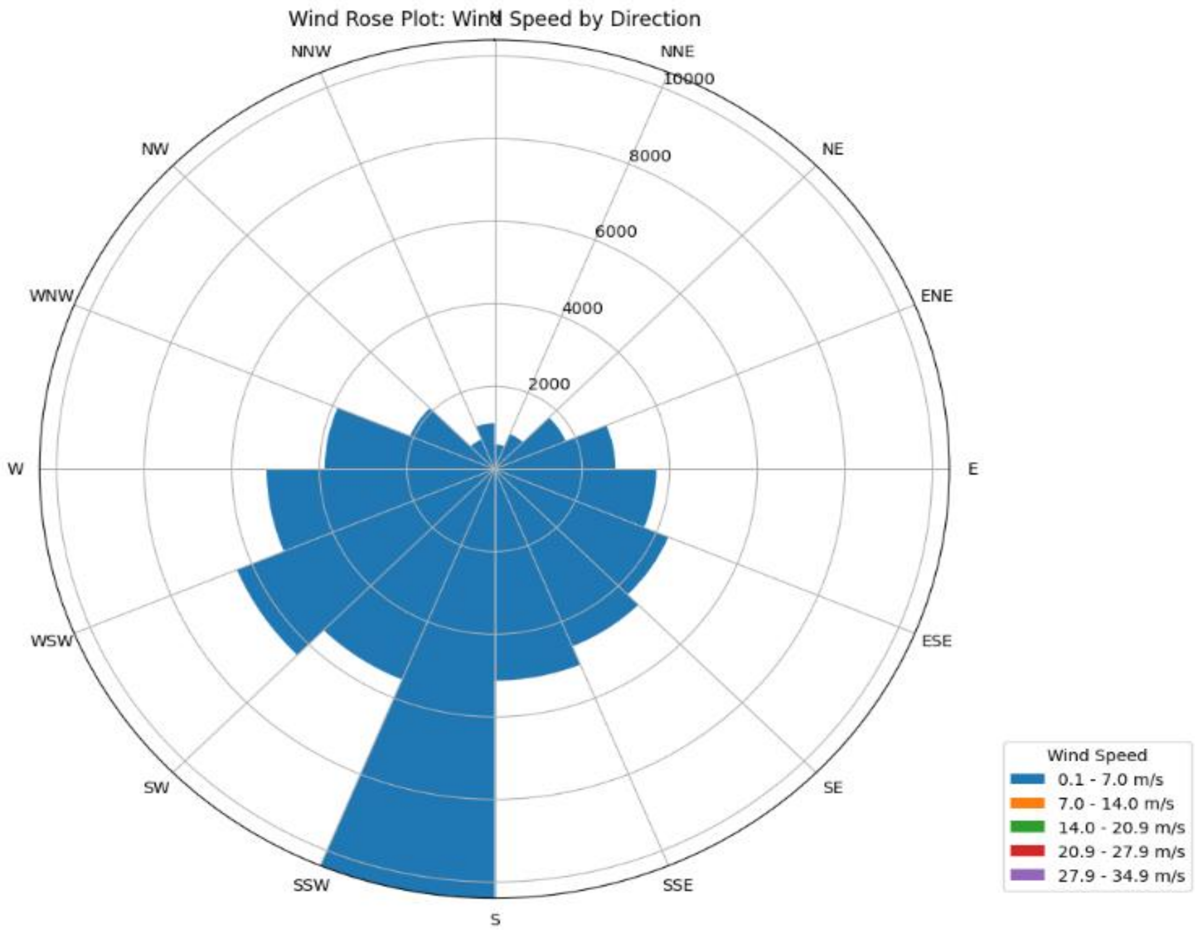
Wind Rose Plot

The wind rose plot represents the distribution of wind speed and wind direction over the observed period. Each segment shows the frequency of wind coming from a particular direction, while the colours indicate different wind speed ranges.

From the plot, it is evident that

- The dominant wind direction is South (S) and South-Southwest (SSW**)**, showing the highest frequency of wind occurrence.
- Moderate wind speeds (7.0–14.0 m/s**)** are most common across all directions.
- Higher wind speeds (14.0–20.9 m/s) occur occasionally, mainly from southern and south-western directions.
- Very high wind speeds (>20.9 m/s) are rare, indicating relatively stable wind conditions.
- Northern and north-eastern directions contribute the least wind activity.

Overall, the wind rose highlights that wind patterns are directionally consistent, with stronger and more frequent winds coming from the southern sector, which is useful for applications such as weather analysis, wind energy planning, and environmental studies.



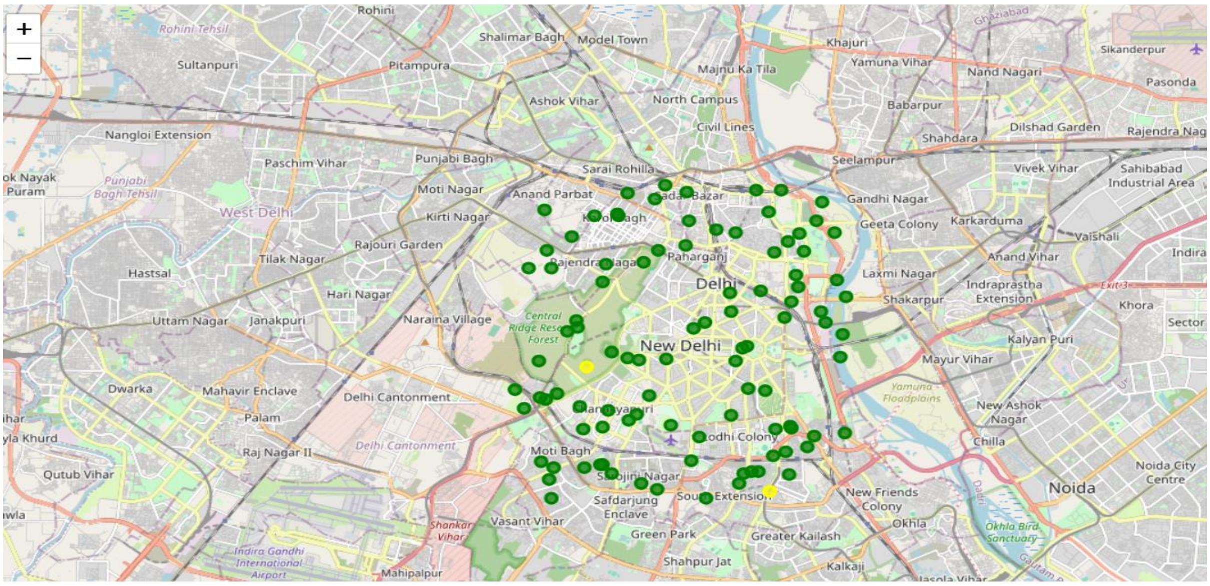
Geographical Pollution Map

The geographical pollution map visualizes the spatial distribution of air pollution monitoring locations across Delhi. Each marker represents a monitoring point, highlighting areas with varying pollution intensity and coverage.

Key observations from the map include

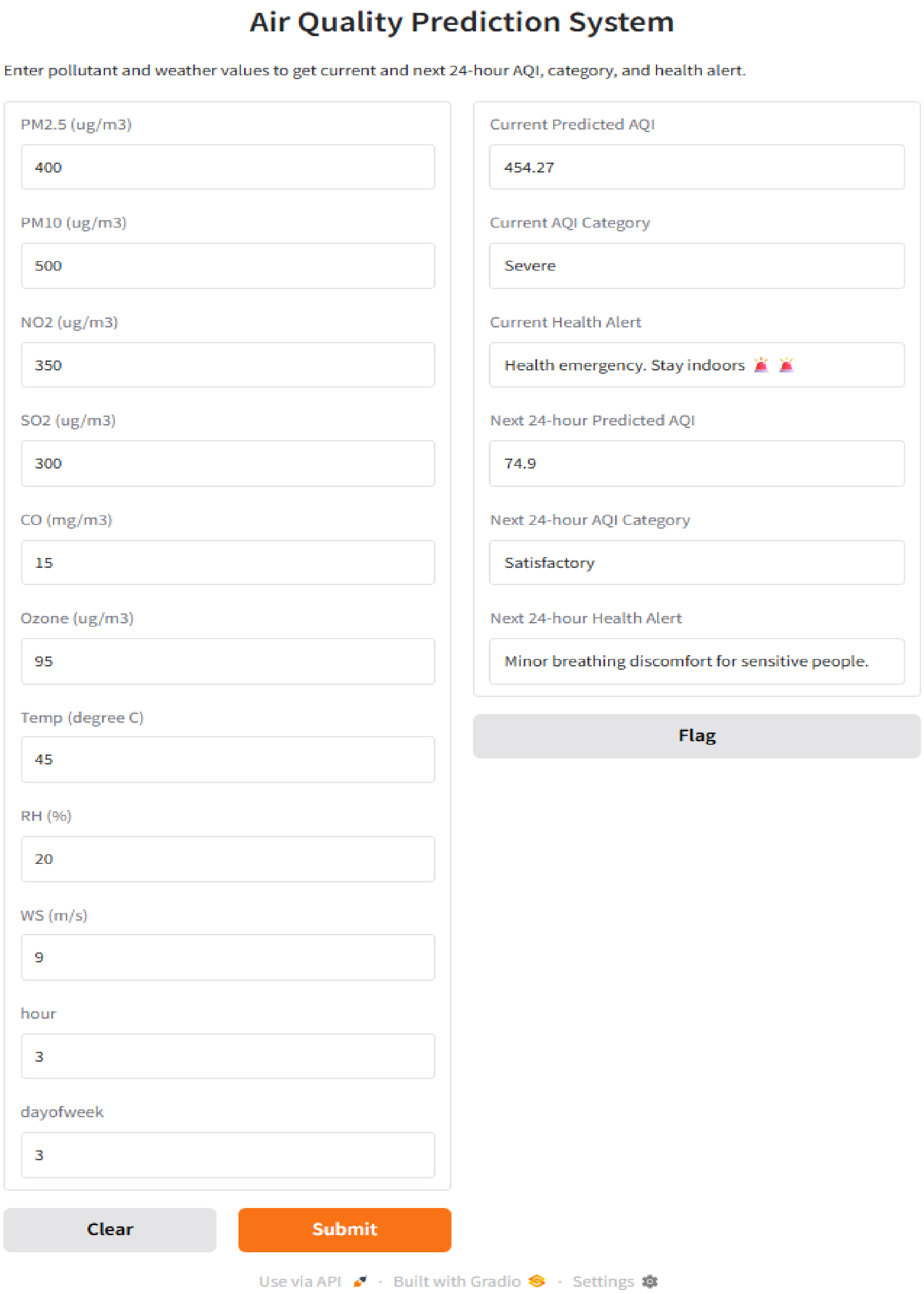
- High concentration of monitoring points in Central and East Delhi, indicating dense urban activity and traffic congestion.
- Pollution hotspots are visible near commercial hubs, major road networks, and industrial zones.
- South and South-East Delhi show moderate pollution spread, influenced by residential density and vehicular emissions.
- Comparatively fewer monitoring points are observed in peripheral regions, suggesting lower activity or limited data coverage.
- The map helps identify pollution-prone zones and supports targeted policy actions and environmental planning.

Overall, this geographical visualization provides a clear understanding of pollution patterns, enabling effective decision-making for air quality management, public health analysis, and urban planning.



Layouts

The displayed interface represents the final operational stage of the Air Quality Prediction System, where user-provided pollutant and meteorological inputs are processed to generate both current and next 24-hour AQI predictions. The layout is designed to clearly separate the input section from the output results to ensure usability and clarity. On the left side of the interface, users enter pollutant concentrations such as PM2.5, PM10, NO_2, SO_2, CO, and Ozone, along with meteorological parameters like temperature and other environmental factors. These inputs serve as independent variables for the trained machine learning model. Once the values are submitted, the system processes the data using the trained Random Forest Regression model and generates multiple outputs displayed on the right-hand panel. The first output shown is the Current Predicted AQI, which is a numerical representation of the present air quality condition based on the entered pollutant levels. In the example shown, the predicted AQI value is approximately 214.71, which falls under the “Poor” category. This classification is automatically determined using predefined AQI breakpoints, making the result easier for users to interpret. Below the AQI value, the interface displays the Current AQI Category, which translates the numerical AQI into a descriptive pollution level. This enhances readability and allows non-technical users to quickly understand the air quality status without interpreting raw numbers. Additionally, a Current Health Alert message is provided, offering precautionary guidance based on the predicted pollution level. For instance, the alert indicates that breathing discomfort is likely and prolonged exposure should be avoided. This feature transforms the system from a purely predictive model into a practical public health advisory tool. The interface also includes a Next 24-hour Predicted AQI value, which forecasts future air quality conditions using learned temporal relationships in the dataset. In the example, the predicted future AQI is significantly lower (approximately 98.92), categorized as “Satisfactory.” This indicates an expected improvement in air quality conditions. A corresponding health alert is also displayed for the predicted future AQI, providing early awareness and helping users plan activities accordingly. The overall structure of the output page ensures that complex machine learning predictions are presented in a simple, user-friendly, and actionable manner. By combining numerical predictions, categorical interpretation, and health advisories, the system enhances usability, interpretability, and real-world applicability. The interface design implemented using Gradio supports interactive execution and ensures that the prediction results are displayed dynamically and clearly.



9.3 Accuracy

To assess the effectiveness and reliability of the developed Air Quality Prediction System, appropriate evaluation metrics were applied for both regression and classification tasks. Since the project involves predicting continuous AQI values as well as categorical AQI levels, different performance measures were used accordingly. For the AQI value prediction task, which is a regression problem, the performance of the Random Forest Regressor model was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score. Mean Absolute Error measures the average absolute difference between actual AQI values and predicted AQI values, providing a direct interpretation of prediction error in AQI units. A lower MAE indicates that the model's predictions are closer to the true values. Mean Squared Error measures the average squared difference between predicted and actual AQI values. Since errors are squared, larger deviations are penalized more heavily, making MSE sensitive to outliers. A lower MSE indicates better predictive performance and stability. The R^2 Score, also known as the coefficient of determination, measures how well the independent variables explain the variability in the AQI values. An R^2 value closer to 1 indicates that the model successfully captures most of the variance in air quality data, demonstrating strong predictive capability. For the AQI category prediction task, which is a classification problem, model performance was evaluated using Accuracy. Accuracy represents the proportion of correctly classified AQI categories out of the total predictions. A high accuracy value indicates that the model effectively assigns pollution levels such as Good, Moderate, or Poor based on predicted AQI values. This metric helps assess how reliably the system translates numerical predictions into meaningful pollution categories. By combining regression and classification evaluation metrics, the system's predictive strength was thoroughly validated. The use of MAE, MSE, and R^2 ensures that numerical AQI predictions are precise and stable, while classification accuracy confirms that the pollution category outputs are reliable and interpretable. This comprehensive evaluation approach strengthens the credibility and robustness of the proposed Air Quality Prediction System.

For AQI category classification, a classification report was generated including precision, recall, F1-score, and overall accuracy. The model achieved an overall accuracy of 1.00 (100%), indicating that all test samples were correctly classified into their respective AQI categories. The confusion matrix further confirms perfect classification performance, as all predicted values lie along the diagonal with zero misclassifications. Precision, recall, and F1-score for each category — Good, Satisfactory, Moderate, Poor, Very Poor, and Severe — are all equal to 1.00. This indicates that the classifier correctly identified every instance of each class without false positives or false negatives. The macro average and weighted average values are also 1.00, demonstrating balanced performance across both majority and minority classes. However, it is important to note that extremely high classification accuracy may indicate that AQI categories were directly derived from calculated AQI values, which makes the classification task relatively straightforward.

```
              precision    recall  f1-score   support

        Good       1.00      1.00      1.00      7471
    Moderate       1.00      1.00      1.00       953
        Poor       1.00      1.00      1.00        88
Satisfactory       1.00      1.00      1.00      3306
      Severe       1.00      1.00      1.00         2
   Very Poor       1.00      1.00      1.00        10

    accuracy                           1.00     11830
   macro avg       1.00      1.00      1.00     11830
weighted avg       1.00      1.00      1.00     11830

[[7471    0    0    0    0    0]
 [   0  953    0    0    0    0]
 [   0    0   88    0    0    0]
 [   0    0    0 3306    0    0]
 [   0    0    0    0    2    0]
 [   0    0    0    0    0   10]]
```

For the numerical AQI value prediction, regression evaluation metrics were used. The Mean Absolute Error (MAE) obtained is 0.0067, which indicates that the average absolute difference between the predicted AQI and actual AQI values is extremely small. This suggests that the model predictions are nearly identical to the true values. The Root Mean Squared Error (RMSE) is 0.2801, which is also very low, showing that large deviations are minimal and the model produces highly stable predictions. The R^2 Score is 0.99995, which means that approximately 99.995% of the variance in AQI values is explained by the model. This demonstrates an almost perfect fit between the predicted and actual values.

```
MAE: 0.006745529763637095
RMSE: 0.280133438639118
R2 Score: 0.9999497258113634
```

Overall, the evaluation results indicate that the Random Forest-based Air Quality Prediction System performs with exceptionally high accuracy for both AQI classification and regression tasks. The extremely low error values and near-perfect R^2 score demonstrate the model’s strong capability in capturing the relationship between pollutant concentrations and AQI levels. These results confirm the robustness and reliability of the proposed system in predicting current air quality conditions.

CONCLUSION

10. Conclusion

This project successfully developed a machine learning-based Air Quality Prediction System capable of estimating both the current AQI and the next 24-hour AQI using pollutant and meteorological parameters. The system utilized historical air quality data from the AP001 monitoring station and applied data preprocessing techniques to ensure data quality. A Random Forest Regression model was implemented to predict AQI values, which were further categorized into standard air quality levels such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe. The model achieved excellent predictive performance with very low error values and a high R^2 score, demonstrating its ability to accurately capture the relationship between pollutant concentrations and air quality levels. In addition, an interactive user interface was developed using Gradio to allow users to easily input environmental parameters and obtain AQI predictions along with health advisory messages. Overall, the proposed system provides a reliable, data-driven approach for air quality monitoring and short-term forecasting. It has the potential to support environmental awareness and assist in decision-making related to pollution control and public health protection.

FUTURE ENHANCEMENTS

11. Future Enhancements

Although the current Air Quality Prediction System achieves high performance, it can be further developed into a complete production-level data science solution. Currently, the system focuses mainly on model training and prediction. Future improvements can extend it across the entire data science lifecycle, including automated data engineering, advanced modeling, explainability, monitoring, and deployment. An automated ETL pipeline can be implemented to continuously collect and preprocess real-time air quality and meteorological data from APIs, IoT sensors, and satellite sources. This would transform the model from a static dataset-based system into a scalable, real-time learning solution. Advanced feature engineering techniques such as lag variables, rolling averages, seasonal indicators, and interaction features can improve predictive performance. Multiple models like XGBoost, LightGBM, Support Vector Regression, and LSTM can be evaluated using cross-validation, along with hyperparameter tuning methods such as Grid Search and Bayesian Optimization to ensure robustness and generalization. Explainable AI techniques like SHAP can be integrated to interpret model predictions and identify the impact of pollutants such as PM2.5 and NO_2 on AQI levels. Additionally, model monitoring and drift detection mechanisms can be implemented to maintain long-term reliability. From a deployment perspective, the system can be converted into a cloud-based API using Docker and CI/CD pipelines, along with an interactive dashboard for visualization. The project can further evolve into prescriptive analytics by simulating pollutant reduction strategies and recommending emission control measures. Overall, these enhancements would transform the system into an end-to-end, industry-ready data science solution with strong practical relevance.

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